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The simulation of SST, sea ice extents and ocean heat transports in a version of the Hadley Centre coupled model without flux adjustments

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Abstract Results are presented from a new version of the Hadley Centre coupled model (HadCM3) that does not require flux adjustments to prevent large climate drifts in the simulation. The model has both an improved atmosphere and ocean component. In particular, the ocean has a $1.25^\circ \times 1.25^\circ$ degree horizontal resolution and leads to a considerably improved simulation of ocean heat transports compared to earlier versions with a coarser resolution ocean component. The model does not have any spin up procedure prior to coupling and the simulation has been run for over 400 years starting from observed initial conditions. The sea surface temperature (SST) and sea ice simulation are shown to be stable and realistic. The trend in global mean SST is less than 0.009°C per century. In part, the improved simulation is a consequence of a greater compatibility of the atmosphere and ocean model heat budgets. The atmospheric model surface heat and momentum budget are evaluated by comparing with climatological ship-based estimates. Similarly the ocean model simulation of poleward heat transports is compared with direct ship-based observations for a number of sections across the globe. Despite the limitations of the observed datasets, it is shown that the coupled model is able to reproduce many aspects of the observed heat budget.

1 Introduction

Coupled ocean-atmosphere general circulations models (OAGCMs) have become a valuable tool in attempting to understand and predict climate change (Houghton et al. 1996). One of the major drawbacks of these models

has been the large climate drifts that occur when they are used to simulate the current climate. In many models these drifts have been alleviated by the use of flux adjustments (Sausen et al. 1988; Manabe et al. 1991; Johns et al. 1997). Over the past few years a number of developments have been made to the Hadley Centre global coupled model that is used in the investigation of anthropogenic climate change. Recent improvements to both the atmosphere and ocean models have gone a long way towards reducing the model climate drift, and thereby obviated the need for flux adjustment. Other models have also recently demonstrated the ability to produce stable climate simulations without flux adjustments (Barthelet et al. 1998; Boville and Gent 1998).

This work describes results from a simulation of pre-industrial climate using the latest version of the Hadley Centre coupled model (HadCM3), which does not use flux adjustments. We concentrate on the analysis of the mean simulation of sea surface temperature (SST), sea ice extents, surface heat and momentum fluxes and ocean heat transports, as these are some of the key climate variables for which, in the past, it has been difficult to obtain a realistic and stable simulation.

It is important to establish that the model realises a stable climate for sound physical reasons. In particular, we consider the balance of heating terms and therefore concentrate on the comparison with available observations of both the atmospheric model simulation of surface heat fluxes and the ocean model simulation of heat transports. The lack of salinity drifts are equally important in maintaining a stable climate simulation and, as discussed briefly in a later section, the salinity drifts in the model over 400 years do not lead to major changes in the ocean circulation. The fresh water budget and the ocean salinity simulation will be discussed in detail in a separate paper.

Section 2 discusses climate drift in coupled models and provides the framework for the structure of the rest of the study. In Sect. 3 the various components of the coupled model are described. Section 4 outlines the initialisation of the model and the convergence of

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the coupled model state. The simulation of SST and sea ice extents in the coupled model are discussed in Sect. 5. Section 6 considers the surface fluxes of heat and momentum from a simulation of the atmospheric model using observed SSTs. These are compared with estimates of these fluxes from climatologies based primarily on ship observations. The ocean heat transports in the model are discussed in Sect. 7 and, finally, the results are summarised in Sect. 8.

2 Climate drift in coupled models

The atmosphere and ocean interact via the interface at the sea surface and the SST and sea ice extents are therefore crucial predictands in a coupled model simulation. The simulation of ocean temperatures and sea ice extents also depends critically on ocean heat transports. In any ocean region, in the long term annual mean, the net surface heat exchange must be balanced by the ocean advection of heat into (or out of) the region. In the zonal average this is expressed by the simplified equation for the depth and zonally averaged ocean potential temperature:

$$\rho_0 C_p H \frac{\partial \langle \theta \rangle}{\partial t} + \frac{1}{a \cos \phi} \frac{\partial}{\partial \phi} (F_\theta) = Q_0 \quad (1)$$

$$F_\theta = \rho_0 C_p H \langle v \theta \rangle \cos \phi$$

where $\langle \rangle$ denotes a zonal and depth average, Q_0 is the zonal mean heating at the surface, F_θ is the flux of heat across the latitude ϕ and H the ocean depth. Other variables are standard notation. Diffusion effects have been ignored in Eq. (1) since in most regions they are not a major contribution to the model zonal mean heat budget (see Fig. 17). It follows from Eq. (1) that there will be a net drift in the depth averaged zonal mean potential temperature (i.e. $\partial \langle \theta \rangle / \partial t \neq 0$) if the ocean meridional advection and net surface heating terms do not balance. It is not a simple matter to relate the vertically integrated temperature changes to SST changes, Eq. (1) being only applicable to the vertical mean temperature, and within the vertical column there may be compensating temperature drifts. Equation (1) does, however, provide a very basic framework for discussing climate drift of ocean temperatures in a coupled model.

It is apparent that if the ocean heat transports implied by the atmospheric model surface fluxes are significantly different to the actual ocean model heat transports, then the coupled model ocean temperatures must drift to enable a new balance to be established. If a stable equilibrium exists then the SST and sea ice extents must drift in such a way as to bring the model implied heat transports to be in balance with the ocean heat transports. (Here the terminology ‘implied’ heat transports will be used for transports determined from the integration of air sea fluxes and ‘actual’ heat transports those determined either from direct ocean observations or from the calculation of the temperature advection terms in the ocean model.) Weaver and Hughes (1996)

demonstrated that in a model with climate drift, a ‘minimum’ flux adjustment could be applied which simply ensured the flux balance discussed in a zonal mean sense. Recently Boville and Gent (1998) have, in part, attributed their success in obtaining a realistic un-flux adjusted coupled simulation to the compatibility of the actual and implied heat fluxes (Bryan 1998).

The heat transports in the ocean model depend upon the full three dimensional temperature, salinity and current structure. In particular, salinity is important because of its effects, via the density, on the general circulation. It can also be important locally at the surface in determining the mixed layer structure and thereby affecting the mixed layer, and SST, response to surface heat forcing. Even if the global-average salinity is constant, a poor simulation of the precipitation, evaporation, runoff and ice melt and freezing can all lead to local salinity drifts. The long term ocean response to a poor simulation of surface water fluxes can be potentially more serious than that for heat fluxes, because there is no feedback on the fluxes from the surface salinity as there is from surface temperature. In extreme cases, over-freshening of the surface at high latitudes could lead to a significant reduction in the thermohaline circulation. In HadCM3 systematic errors in the fresh water budget do not lead to large changes in circulation. We concentrate on the analysis of the heat budget.

Earlier versions of the Hadley Centre coupled model (‘UKTR’ and ‘HadCM2’), which required flux adjustment, certainly showed an imbalance between the implied and actual ocean heat transports. In these earlier versions, SSTs were constrained to be close to reality by application of a flux adjustment in the surface heating term. Figure 1 shows the ocean heat transport from two coupled integrations with the earlier versions of the Hadley Centre model. Details of HadCM2 can be found in Johns et al. (1997) and UKTR in Murphy (1995). Both the implied heat transports calculated from the atmospheric model surface fluxes and the actual ocean heat transports from the ocean model are shown in Fig. 1. For comparison, ocean heat transport estimates using the Da Silva et al. (1994) flux climatology (hereafter referred to as DS) are also included and, following DS, we assume a transport of 0.1 PW at 65°N. The calculations based on the AGCM fluxes for both models show the same overall pattern as the climatology in the Northern Hemisphere. The HadCM2 fluxes are clearly closer to the DS values. The differences in the Southern Hemisphere will be discussed further in Sect. 6. Figure 1 also clearly shows that the ocean models used in HadCM2 and UKTR, when integrated in flux adjusted mode, underestimate the ocean heat transports by nearly a factor of two. The flux adjustment term essentially makes up the difference. When the HadCM2 model is integrated without flux adjustments (Gregory and Mitchell 1997) there is a large SST drift as the coupled model attempts to reach an unrealistic balanced state.

These factors point to the fact that a realistic simulation of both the surface heat fluxes and the ocean heat

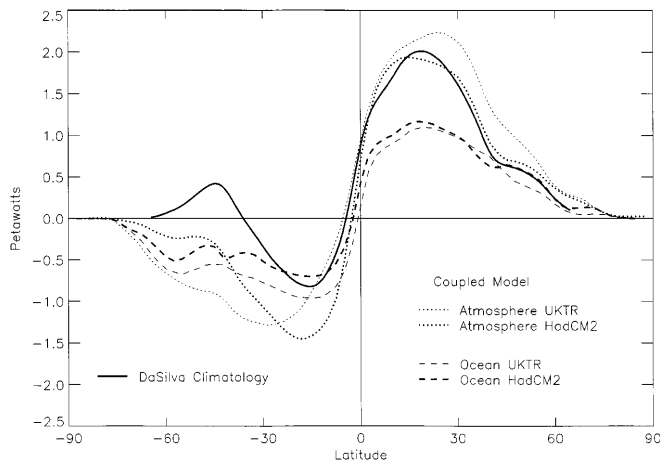


Fig. 1 The zonally averaged annual mean northward heat transport as a function of latitude calculated from the atmosphere and ocean components of previous flux adjusted versions of the Hadley Centre model (units PW). Climatological estimates based upon the Da Silva et al. (1994) compilation of surface fluxes are also included

transports is necessary to reduce climate drift in the surface temperature and sea ice simulation. It is therefore important to evaluate the heat budget in both component models rather than simply to assess the coupled model SST and sea ice simulation. An initial comparison of HadCM3 with available observations will be presented here. Experience with the un-flux adjusted HadCM3 model has also shown that, away from the equator and strong current regions, the seasonal simulation of SST is strongly dependent on the surface fluxes. This dependence is in agreement with the observations (Gill and Niiler 1973).

3 Model description

3.1 The atmospheric component

The atmospheric model component in HadCM3 is a version of the UKMO unified forecast and climate model run with a horizontal grid spacing of $2.5^\circ \times 3.75^\circ$ and 19 vertical levels using a hybrid vertical co-ordinate. The time step is 30 min. The performance of this model version in a simulation forced by observed SSTs is described in Pope et al. (submitted 2000).

Some of the major changes in this model over the previous version used in coupled simulations (Johns et al. 1997) are now described.

1. A new radiation scheme has been included which has 6 (8) spectral bands in the shortwave (longwave) and represents the effects of minor trace gases as well as CO_2 , H_2O and O_3 . (Edwards and Slingo 1996). A parametrisation of a simple background aerosol climatology (Cusack et al. 1998) is also included.

2. The convection scheme has been improved by adding a parametrisation of the direct impact of convection on momentum (Gregory et al. 1997).

3. A new land surface scheme (Cox et al. 1998) includes the representation of the freezing and melting of soil moisture and the formulation of evaporation includes the dependence of stomatal resistance on temperature, vapour pressure and CO_2 .

4. A parametrisation of orographic drag (Milton and Wilson 1996) and a new gravity wave drag scheme including anisotropy of

orography, high drag states and flow blocking, and trapped lee waves have been included (Gregory et al. 1998).

5. The partitioning of mixed phase clouds into ice and water has been changed from 0 to -15°C to 0 to -9°C (Gregory and Morris 1996) based on evidence from observational data (Moss and Johnson 1994). A parametrisation of the effective radius of cloud droplets as a function of cloud water content and droplet number concentration is also included (Martin et al. 1994). Several parameters in the layer cloud scheme (Smith 1990) have been altered. Cloud cover forms when the standard deviation of the distribution of total water content in a grid box goes above a critical relative humidity (RH_{crit}). RH_{crit} is a constant value at each level of the model. In HadCM3 this has been changed from 0.85 to 0.7 to improve the top-of-the-atmosphere (TOA) radiation balance. The equation for liquid precipitation includes a threshold value of the total water content, C_w , below which water does not precipitate. The value of this threshold is different over land and ocean to represent the different amounts of cloud condensation nuclei. In HadCM3, these values were reduced from 8×10^{-4} to 2×10^{-4} over land and from 2×10^{-4} to 0.5×10^{-4} over sea. In combination, these changes improved the net TOA radiative fluxes when compared with Earth Radiation Budget Experiment (ERBE) data, especially over northern mid-latitude oceans.

6. In HadCM3 there were some significant revisions to the boundary layer mixing used in HadCM2; most importantly the removal of a non-local mixing scheme (Smith 1993), the inclusion of which was found to degrade the transport and sink of aerosols.

In addition to these major model improvements, there have been a number of minor changes. These are documented in Pope et al. (1999).

3.2 The ocean component

The ocean model has been specifically developed over a number of years for use in coupled climate simulations. The version used in the simulation described is substantially different to that previously employed in simulations with the Hadley Centre coupled climate model. The basic model, which previously had a $2.5^\circ \times 3.75^\circ$ horizontal resolution ocean component, is described in Johns et al. (1997). Here the modifications made in the current version will be focused upon.

The ocean component is a 20 level version of the Cox (1984) model on a $1.25^\circ \times 1.25^\circ$ latitude-longitude grid. There are six ocean grid boxes to each atmosphere model grid box and each high latitude ocean grid box can have partial sea ice cover. The vertical levels are distributed to provide enhanced resolution near to the ocean surface and are the same as those in the previous coarser horizontal resolution version of the model (Johns et al. 1997). The topography was taken from the ETOPO5 (1988) $1/12^\circ$ resolution dataset and interpolated onto the model grid. A simple smoother was applied to remove gridscale noise. Roberts and Wood (1997) showed that Bryan-Cox type models are highly sensitive to the depth of the various channels along the Greenland-Iceland-Scotland ridge. Many of these channels are sub-gridscale and so three routes (one grid point wide on the velocity grid) through the ridge were excavated. The Denmark Strait and Iceland-Faeroes ridge were reduced to 797.9 m (bottom of level 12), while the Faeroe-Scotland ridge was set to 534.7 m (bottom of model level 11). This leads to a long-term mean outflow of approximately 8.5 Sv in the coupled simulation, compared with the observed outflow of around 5–6 Sv (Dickson and Brown 1994). An island is also placed at the North Pole to avoid the polar singularity in the spherical co-ordinate system.

Horizontal eddy mixing of tracers is parameterised using the Visbeck et al. (1997) version of the Gent and McWilliams (GM) (1990) adiabatic thickness diffusion and the Redi (1982) along isopycnal diffusion scheme. The near surface vertical mixing is parameterised by a hybrid approach in which the mixing of tracers is carried out via a Kraus-Turner mixed layer sub-model (Kraus and Turner 1967) and a K-Theory scheme. The convective adjustment is modified in the region of the Denmark Straits and Iceland–

Scotland Ridge to better represent the down slope mixing of the overflow water. The adjustment is modified in this region since previous studies have shown the simulation of the sub-polar gyre to be sensitive to the details of the mixing of the overflow water (Roberts et al. 1996). There is also a parametrisation of the outflow of water from the Mediterranean in which water is partially mixed across the Strait of Gibraltar. A detailed description of these schemes can be found in Appendix A.

Shortwave solar radiation is selectively absorbed with depth using a double exponential decay. Attenuation coefficients applicable to a global uniform Jerlov Type IB are used (Paulson and Simpson 1977).

The sea ice model, which is the same as that used in HadCM2, uses a simple thermodynamic scheme and contains parametrisations of ice drift and leads (Cattle and Crossley 1995). The model description is repeated here for completeness. The thermodynamics of the model is based on the zero-layer model of Semtner (1976). A parametrisation of ice concentration based on that of Hibler (1979) is included. Ice concentration is not allowed to exceed 0.995 in the Arctic and 0.980 in the Antarctic since completely unbroken ice cover is rarely observed in reality even in pack ice. Ice forms predominantly by freezing in the leads; it melts at the surface during summer and at the base throughout the year. Ice depth can be increased by the formation of 'white ice' (Ledley 1985) where the weight of snow forces the ice-snow interface below the water line.

The effect of sea ice formation and melt on ocean salinity is accounted for within the model, assuming a constant salinity of 0.6‰ for sea ice. Sublimation increases ocean salinity, as the salt is assumed to blow into leads, and white ice formation reduces it to account for the salt added in converting snow to ice. Snowfall reduces ocean salinity in leads, and accumulates onto ice, and all rainfall is assumed to reach the ocean through leads.

Surface fluxes over the ice and leads fractions of each grid box, and surface temperatures, are calculated separately within the atmosphere component of the model, assuming a linear temperature profile in the ice. The surface albedo is 0.8 at -10°C and below, and between -10 and 0°C it falls linearly to 0.5. This parametrisation aims to reproduce some of the effect of the ageing of snow, the formation of melt ponds, and the relatively low albedo of bare ice.

A windmixing energy (see Appendix A), used in the mixed layer model, is calculated using the drag coefficient appropriate for leads and weighted by the leads fraction of the grid square. Oceanic heat flux into the base of the ice is related to the temperature difference between the ocean top level and the base of the ice (assumed to be at freezing point of -1.8°C) with a coupling coefficient of $20\text{ W m}^{-2}\text{ K}^{-1}$.

A simple parametrisation of sea ice dynamics based on Bryan (1969) is also included. The windstress is applied to the ocean beneath the ice. The ice thickness, concentration and snow depth are advected using the top layer ocean current, using an upstream advection scheme. Ice rheology is crudely represented by preventing convergence of ice once the ice depth reaches 4 m (Steele et al. 1997). The ice may become deeper than 4 m due to further freezing. Del-squared horizontal diffusion of ice depth is also applied, with a coefficient $2000\text{ m}^2\text{ s}^{-1}$.

3.3 Coastlines and Indonesian through-flow

Although the ocean model has a $1.25^{\circ} \times 1.25^{\circ}$ horizontal grid, for ease of coupling the coastline used in the coupled model is that of the atmospheric model $2.5^{\circ} \times 3.75^{\circ}$ grid. In particular, this means that there is only a very coarse representation of the channels between the Pacific and Indian Oceans in the Indonesian through-flow region. In reality the major gap for barotropic flow from the Pacific to the Indian ocean is between Papua New Guinea (PNG) and the Indonesian islands, with flow between Australia and PNG blocked due to shallow topography. Because of the coarse grid coastline, flow occurred through an unrealistic gap between Indonesia and Asia in a preliminary setup of the model. In addition, the minimum water depth of 140 m assumed in the model permitted too much through-flow. To compensate for the coarse grid, baro-

tropic flow is therefore only allowed in the model described here between PNG and Indonesia.

3.4 Atmosphere-ocean coupling

The models are coupled once per day. The atmospheric model is run with fixed SSTs through the day and the various forcing fluxes are accumulated each atmospheric model time step. At the end of the day these fluxes are passed to the ocean model which is then integrated forwards in time. The updated SSTs and sea ice extents are then passed back to the atmospheric model. As there are six ocean grid points to every atmospheric grid point interpolation and/or averaging is used to transfer fields between the two grids (see Appendix B for details).

River outflow is also included allowing ocean salinity feedbacks via changes over land. Runoff is converted into river outflow using river catchments over land and associated coastal outflow points are defined relative to the model grid. River transport is not modelled explicitly, so runoff is transported instantaneously to the coast.

3.5 Water conservation

The rigid lid in the ocean model means that the fresh water flux at the ocean surface does not lead to changes in the volume of water in the column. In order to close the global salinity budget, the conversion of the surface water flux W , into an equivalent salinity flux F_s , is achieved by $F_s = -WS_0/\rho_0$, where ρ_0 is a constant reference density and S_0 a constant salinity of 35‰.

Using the constant S_0 , instead of the spatially and temporally varying S , means that there is no drift in global mean salinity when there is no net water flux, but the local effect of surface water fluxes will be exaggerated in regions where $S < S_0$, and under-represented when $S > S_0$. The impact of this boundary condition on the model simulation is generally small (Lowe and Gregory 1998).

A small amount of model rainfall runs into internal drainage basins and a small term, equivalent to 0.01 mm day^{-1} , is applied uniformly over the ocean to allow for this. Even with river runoff included, the area-average water flux in HadCM3 coupled model is not zero. Most of the imbalance, equivalent to a loss of 7.3 mm year^{-1} of water averaged over the ocean and a salinity drift of 0.1‰ in 1000 years, comes from the accumulation of snow on the Antarctic and Greenland ice-sheets, 80% on the former. (These rates compare fairly well with those summarised in Houghton et al. 1996 and with other coupled models such as Russell et al. 1995.) In the real world, fresh water is returned to the ocean by the calving of icebergs, a process not represented in the model. The mass balance of the ice sheets is not well known so we therefore assume that accumulation should balance calving for each ice sheet, and apply an appropriate water flux uniformly over the areas of the adjacent oceans where icebergs occur (Harvey 1976; Bigg et al. 1996). Of course, the meltwater of the icebergs should not really be uniformly distributed, but the model is not noticeably sensitive to the distribution, furthermore the current state of observations and modelling does not justify any more elaborate treatment of this relatively small term. The largest local values of the iceberg term are about 0.15 mm day^{-1} .

Over the North Atlantic iceberg region, these additional terms are about 5% of the size of $P-E$. In Baffin Bay and much of the Arctic, they are between 10% and 20% and are also above 10% in limited areas around Antarctica, and in small regions near the coast above 20%. Over the majority of the world ocean, they are less than 1% of the size of $P-E$.

A problem arises with the salinity simulation for small isolated basins, such as the Caspian Sea, and others in the model, such as the Baltic, where there is no mixing across the straits which connect them to the world ocean. In these latter cases, the model would be improved by including exchange across the straits (as done, e.g., by Russell et al. 1995). In HadCM3, the salinity at every point is constrained to remain within the limits of 0 and 40‰ . Some isolated basins reach these limits; keeping them there implies a small

non-conservation of water amounting to 0.2 mm year^{-1} averaged over the ocean.

These issues associated with water conservation and its effects on the salinity drifts in the model were masked in earlier model versions by the use of salinity flux adjustment terms.

4 Coupled model initialisation and convergence

The coupled model was initialised with an atmospheric model state appropriate to mid-September and ocean temperatures and salinities were specified from the Levitus and Boyer (1994) and Levitus et al. (1995) mean climatology for the month of September. Ocean currents were zero at the initial time. Initial sea ice extent, thickness and concentration were taken from a previous model simulation. The coupled model simulation was integrated for over 400 years and most of the results that follow in the subsequent sections show fields averaged over the years 81–120 and years 361–400. It is noteworthy that no prior stand-alone ocean spin-up of the atmosphere and ocean/sea ice models is needed to achieve a balanced initialised state for use in the coupled model.

Comparison of the coupled model simulation with observations is complicated by the pre-industrial trace gases that are used in this simulation of a ‘control’ climate. The model is simulating the pre-industrial period and, as such, data from this time are ideally needed for model evaluation. Such data is not available and more recent data must be used. The model is also initialised with the Levitus and Boyer (1994) and Levitus et al. (1995) temperature and salinity climatology since a pre-industrial initial dataset is also not available. These complications need to be considered in assessing the comparisons that follow. In general, however, comparison of the GISST SSTs for the 1870s and the 1980s shows that the local model drifts discussed in later sections are much larger than the differences expected because of the mis-matching of time periods.

The time series of global mean SST anomaly and top of the atmosphere (TOA) radiative heat flux are shown in Fig. 2. The SST anomaly is defined as the difference between the model SST and the GISST2.2 values (Rayner et al. 1996). The GISST temperatures are used for comparison as they are based on surface ship and satellite observations and therefore have a much better data coverage than the Levitus XBT based climatology. The annual mean GISST and Levitus and Boyer (1994) and Levitus et al. (1995) SST climatologies are in agreement over most regions of the globe to within a few tenths of a degree centigrade. Larger differences of around one degree occur in the Nordic, Barents and Kara Seas and in the Southern Ocean. The differences in the climatology do not affect

the conclusions drawn about the major model systematic SST errors highlighted in later sections.

The pre-industrial trace gases used will lead to the model being approximately 0.3°C colder than the present day GISST temperatures (Houghton et al. 1996). Figure 2 shows that the global SST in the coupled model remains within 0.3°C of climatology over the 400 years of the simulation and there is little sign of a long term systematic drift. Certainly the drifts are much smaller than the observed rate of global warming. Over the 400 years, the global mean model SST drift is -0.009°C per century.

There is considerable interannual and interdecadal variability evident in the time series. The major characteristics of the geographical pattern of simulated SST are also very stable after the first few decades (to be discussed in the next section). The TOA flux is also generally less than 0.5 Wm^{-2} . Over most of the latter part of the run it is negative, indicating a net cooling of the climate system.

The temperature and salinity drifts below the ocean surface are illustrated in Fig. 3a, b. In the upper 1000 m for the first 50 years of the simulation there is a drift in temperatures away from the initial Levitus and Boyer (1994) and Levitus et al. (1995). After this time the drift at these depths becomes considerably smaller with maximum temperature differences 0.5°C too warm at 200 m in the zonal mean. The source of this warming is not clear and is being further investigated. In the deep ocean there is a slow cooling which will have a very long equilibration time scale. The time series of global volume weighted temperature in Fig. 3c shows there is a net cooling trend of 0.02°C per century. This cooling trend is consistent with the net heat loss indicated by the TOA flux in Fig. 2.

The corresponding drift in salinity is shown in Fig. 3b. After 400 years the depth distribution of the salinity drift shows an increase below 1000 m and a freshening above this depth. Over the 400 years of the integration the global average surface salinity freshens by nearly 0.8‰ , although 50% of this drift occurs in the first 100 years. North Atlantic salinity drifts are particularly important because of their potential impact on the thermohaline overturning circulation. In the North Atlantic the surface salinity freshens by approximately 0.2‰ over the 400 years of the integration, whereas at 500 m there is an increase in salinity of 0.6‰ over a shorter period of 50 years. It will be shown later that the overturning cell in the North Atlantic is stable throughout the integration, demonstrating the relatively small impact of these salinity drifts on the circulation.

The time series of global mean sea ice area and volume are shown in Fig. 4a, b. Monthly and annual mean values are shown. The ice area (Fig. 4a) reaches a stable extent in the first few years of the simulation. The ice volume (Fig. 4b) is also stable but it takes a longer time to reach its long term value. In the first 30 years of the simulation there is a net increase in ice volume indicating an increase in the ice thickness away from its initial value (which was specified from a previous coupled simulation). A comparison with observations of the geographical distribution of sea ice in the model simulation is made in Sect. 5.

Overall we see that after 400 years, the global mean upper ocean and sea ice fields have reached a stage of relatively small drift. This capability to start from the observed ocean state (Levitus and Boyer 1994; Levitus et al. 1995) and to maintain this without large drifts over 400 years is a notable success of the model, particularly as no spin-up is required to precondition the coupled model. How much of the cooling in the deep ocean is a consequence of the pre-industrial trace gas concentrations used has not been established. Since a climatology of ocean conditions in pre-industrial times is not available, proxy data and modelling experiments will be necessary to determine this.

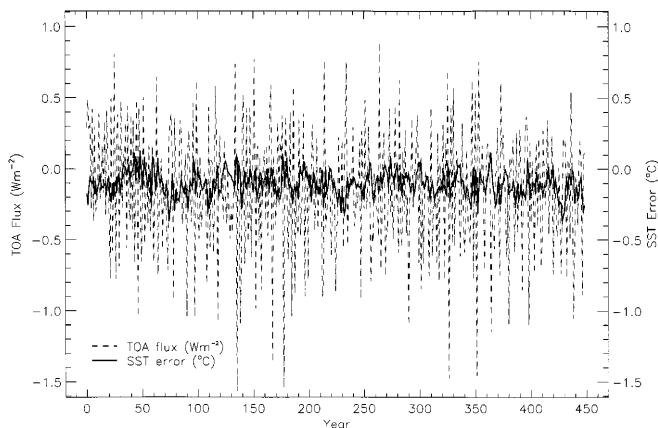


Fig. 2 Time series of the annual global mean SST difference ($^\circ\text{C}$) from the GISST climatology and the annual global mean top of atmosphere flux imbalance (Wm^{-2})

5 Simulation of the sea surface temperature and sea ice extents

The SST and sea ice extents, both in reality and in the coupled model, are the central variables through which the atmosphere/ocean/ice coupling takes place. A realistic simulation of these

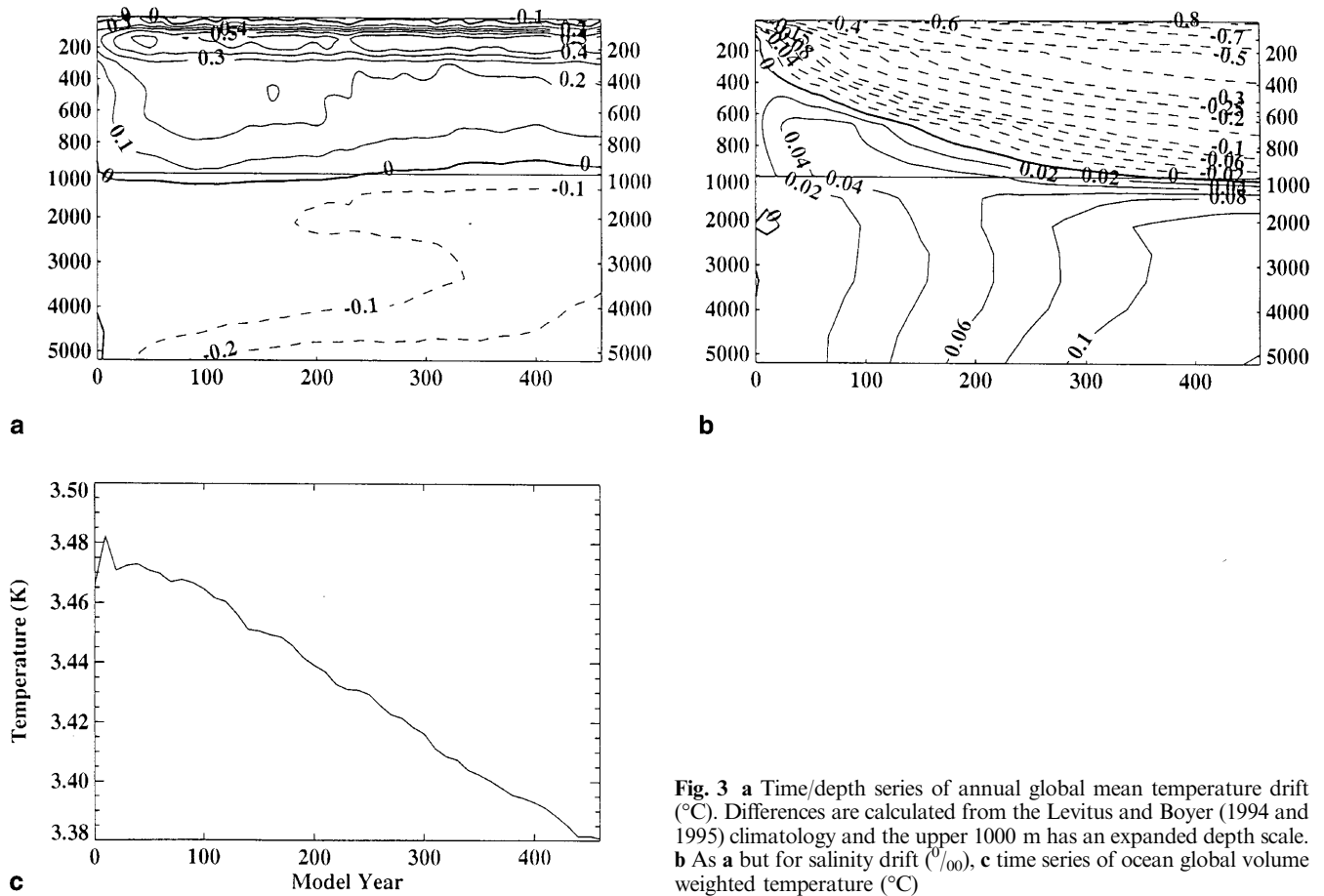
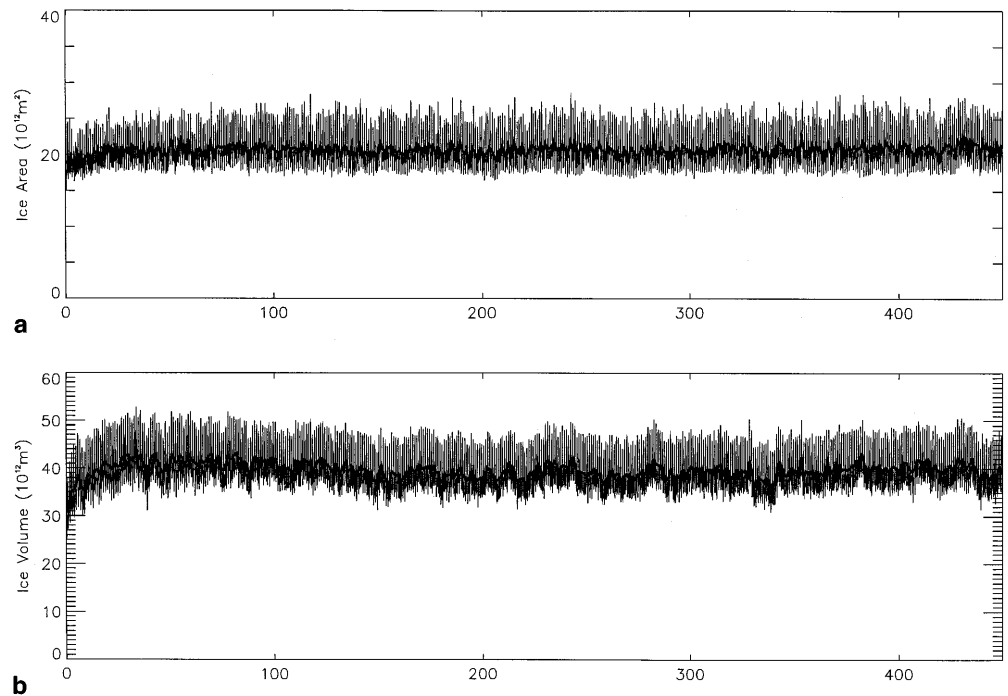


Fig. 3 **a** Time/depth series of annual global mean temperature drift (°C). Differences are calculated from the Levitus and Boyer (1994 and 1995) climatology and the upper 1000 m has an expanded depth scale. **b** As **a** but for salinity drift (‰), **c** time series of ocean global volume weighted temperature (°C)

Fig. 4 **a** Time series of global monthly (*thin line*) and annual (*thick line*) mean sea ice area (10^{12} m^2). **b** Time series of global monthly (*thin line*) and annual (*thick line*) mean sea ice volume (10^{12} m^3)



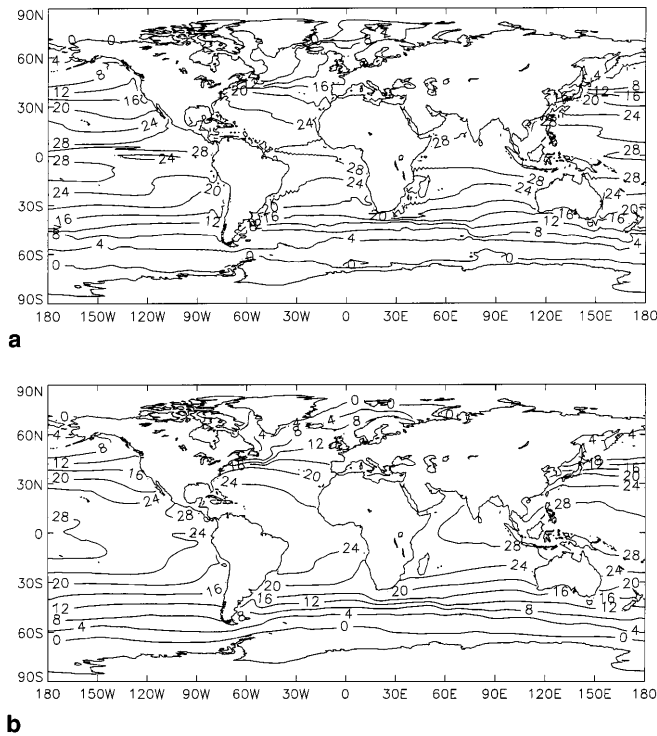


Fig. 5 **a** The coupled model annual mean SSTs (°C) averaged over the years 361–400. **b** The annual mean SST (°C) from the GISST2.2 climatology. In both figures the contour interval is 4 °C

variables is of prime importance in determining the usefulness of the coupled model for climate simulations.

5.1 Sea surface temperature

The simulation of annual mean SST is shown in Fig. 5a averaged over the years 361–400. For comparison the same field from the GISST2.2 observed climatology is shown in Fig. 5b. All of the major observed features in the SST field are reproduced. In particular, the model is able to maintain the sharp horizontal gradients in SST associated with the major ocean currents such as the North Atlantic Current (NAC), Kuroshio and Antarctic Circumpolar Current (ACC). These gradients are maintained in the HadCM3 simulation because of the 1.25° resolution in the ocean model whereas, in earlier non-flux adjusted versions of the coupled model, with a low resolution ocean, these gradients were very diffuse. Of course, in flux adjusted models the gradients are artificially maintained by the flux adjustment terms.

Differences between the model SSTs and the GISST climatology of SST are shown in Fig. 6a, b for the time periods covered by the years 81–120 and 361–400. Over much of the ocean the coupled model after 400 years is simulating the SSTs to within 1 °C of their observed values. There are, however, some notable regions where the differences are considerably larger than 1 °C. A comparison of the SST difference fields for the two time periods shows the pattern is very stable, with the fields being similar in Fig. 6a, b. One notable exception is the greater cooling in the North Atlantic in the later period. The NAC is too zonal in this later period (Fig. 5a) compared to the earlier time, and this leads to the cold SSTs in the region where the NAC in reality turns north. This SST drift does not lead to large change in the Atlantic thermohaline overturning circulation or the poleward heat transports between the two time periods (see later sections and Figs. 19 and 20).

The major systematic error throughout the simulation is the cooling in the North Pacific, where the SSTs are too cold by more than 3 °C over a large region. This cooling establishes itself in the first decade of the coupled simulation and its pattern and

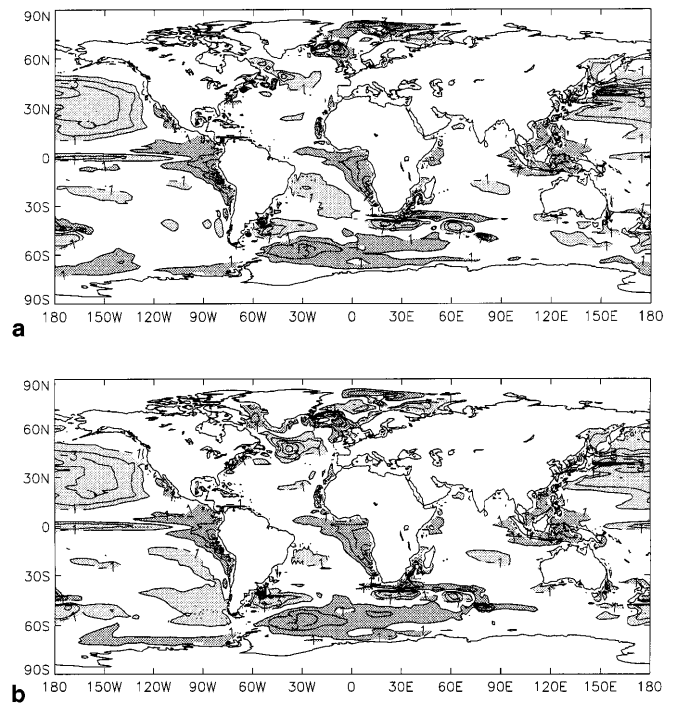


Fig. 6 **a** The model minus climatology annual mean SST (°C) difference averaged over the years 81–120. The contour interval is 1 °C. **b** As **a** but for the model years 361–400

magnitude remain essentially constant throughout the integration. Analysis of the cooling has pointed most strongly to the poor simulation of surface heat fluxes in this region, combined with the very shallow summer-time mixed layer depths occurring in the North Pacific (in the model and in reality). There is also a local maxima in the cooling in the region of the Kuroshio, which separates from the coast too far south in the model. As the Kuroshio and its extension are associated with high horizontal SST gradients, shifts in their position lead to large local SST errors.

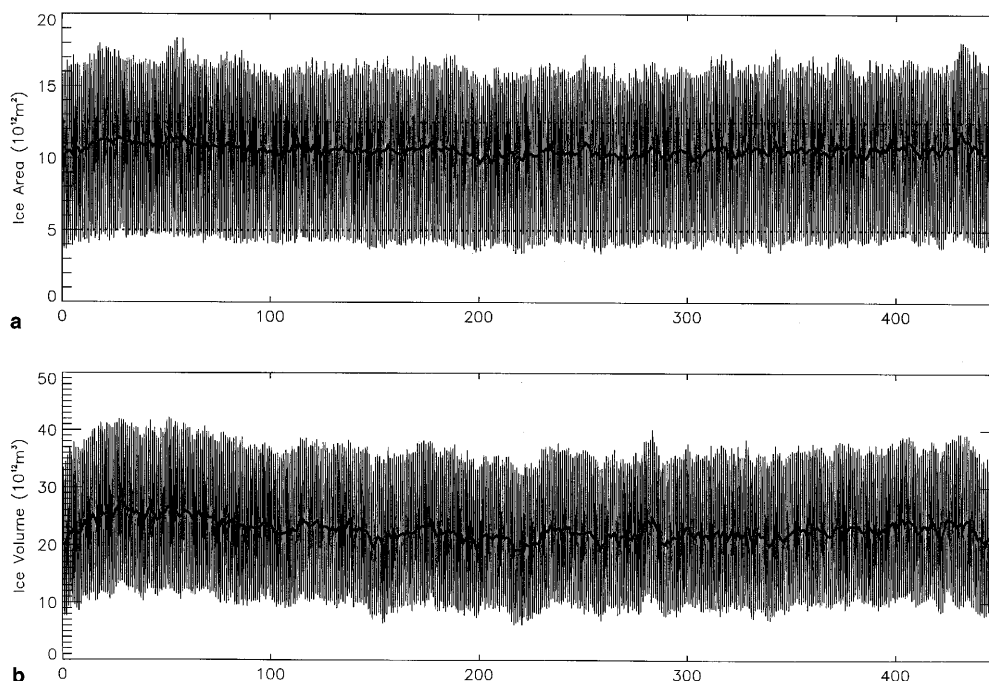
The North Atlantic is reasonably well represented by the coupled model, other than some large local errors associated with shifts in high gradient regions. The tracking of the North Atlantic current shows to some extent the northward turn in the isotherms to the east of the Grand Banks but this is considerably underestimated compared to reality, especially in the later time period.

The model equatorial central east Pacific is too cold, whereas south of the equator the model is too warm in the eastern tropical Pacific and Atlantic. This is to be expected from the lack of simulated stratocumulus cloud in the atmospheric model which will be discussed in Sect. 6. The region off the Californian coast is also too warm, probably for the same reason. The too cold equatorial SSTs are consistent with the excessive equatorial easterlies simulated in the atmosphere-only simulation described later.

The final major error is the warming in the Southern Ocean. There are a number of complex processes that determine the SST in this region. In addition to the surface heat fluxes and advection by the ACC, the seasonal mixing of surface waters with the warmer underlying Circumpolar Deep Water (CDW) also effect the SST. In the south, the ocean is partly covered by ice in winter and the heat used in melting the ice in spring and summer also contributes to the mixed layer heat budget. The model has excessive shortwave fluxes in this region, associated with too little cloud, and this will contribute to the warming. However, sensitivity experiments have shown that the SST warming is also affected by the details of the vertical mixing and the heat lost from the ocean in melting ice.

The peak SST errors in the Southern Ocean are in regions of tight SST gradient and are associated with the incorrect positioning of these gradients (compare the SST fields in Fig. 5a, b).

Fig. 7 **a** Time series of Northern Hemisphere monthly (*thin line*) and annual (*thick line*) mean sea ice area (10^{12} m^2). The observed seasonal extents (*dashed lines*) are from SSM/I data (NSIDC 1989). **b** Time series of Northern Hemisphere monthly (*thin line*) and annual (*thick line*) mean sea ice volume (10^{12} m^3)



These are also recognised as regions of strong topographic steering of the ACC, and the peak SST errors may well be associated with the incorrect steering of the current in the model. A parallel simulation using an ocean model with 40 vertical levels (the version described here has 20 levels) indeed led to some reduction in the peak errors in the Southern Ocean. In the 40 level model the near bottom resolution was doubled leading to a better representation of topographic steering.

One region of particular note is immediately to the south of the Cape of Good Hope in Southern Africa where there is a significant SST error in Fig. 6a, b. This region is known to be high in eddy activity and it has been suggested that there is a substantial eddy heat flux through this area (Thompson et al. 1997). The lack of eddy resolution may be particularly important in this region.

The overall SST error pattern described above is similar in many aspects to that found in both the NCAR CSM1 (Boville and Gent 1998) and ARPEGE T42/OPAICE (Barthelet et al. 1998) non-flux adjusted coupled models. The similarity between the models suggests that the regions described above may be difficult to simulate because of the physical processes governing the SST in these areas, or that the models described have common errors in the simulation. Overall there is a vast improvement in the SST simulation compared to the earlier non-flux adjusted version of the Hadley Centre model (HadCM2) with a low resolution ocean component (Gregory and Mitchell 1997).

5.2 Sea ice

Because of the importance in climate change of the insulation of the ocean surface by ice cover and the ice-albedo feedback, a realistic simulation of sea ice distributions is of critical importance in the coupled model. Time series of sea ice area and volume for both hemispheres throughout the coupled simulation are shown in Figs. 7 and 8. The observed SSM/I (Special Sensor Microwave Imager, NSIDC 1989) maximum and minimum extents are also marked on the figures. In the Arctic the sea ice extents are over-estimated at the maxima when compared to the SSM/I data (Fig. 7a). The seasonal minimum is simulated very well. The reason for this discrepancy in maximum extent can be understood by looking at the modelled geographical distribution of sea ice. Figure 9a shows the simulated annual mean ice concentrations for the Northern Hemisphere for years 361–400. The satellite estimates

from SSM/I are also shown in Fig. 9b. It can be seen by comparing Fig. 9a and b, and the seasonal cycle of Northern Hemisphere ice area (Fig. 10a), that the winter ice extends too far into the North Pacific and that the Barents Sea remains ice covered throughout the year. These contribute to the overestimation of the ice extent maxima. The Northern Hemisphere ice volume (Fig. 7b) initially increases and then stabilises after about 30 years.

The Northern Hemisphere sea ice simulation is much improved over that obtained with un-flux adjusted earlier versions of the Hadley Centre model which employed a low resolution ocean component. In these earlier versions the ice extents were too extensive, reaching south of Iceland in winter. One of the major impacts of using the $1.25^\circ \times 1.25^\circ$ ocean model is a much improved simulation of the North Atlantic Current which advects warmer water into the Norwegian Sea and therefore reduces ice cover in this region. Once this warmer water has entered the basin it should, in reality, flow northwards in the Norwegian Current and then split, one component heading northwards as the West Spitsbergen current, another flowing eastwards into the Barents Sea (Pfirman et al. 1994). In the model almost the entire flow is northwards, with very little of this water getting onto the continental shelf. This failure is partly due to the model's lack of Svalbard as an island. Sensitivity studies have shown that including the island improves the simulation in the region. As already described, without the warm North Atlantic water reaching the Barents Sea, sea ice is too extensive in this region.

The Southern Hemisphere ice extents are simulated well compared to climatology, both in the seasonal variation (Figs. 8a and 10b) and in the mean distribution (Fig. 11a, b). The mean seasonal cycle depicted in Fig. 10b shows the southern hemisphere winter-time sea ice extents to be too large in the model. There is an adjustment time of around 100 years over which the Antarctic sea ice volume increases towards its long term stable value (Fig. 8b).

The general features of the ice motion (not shown) are reasonably well simulated compared with Emery et al. (1997). Advection on the east Greenland coast has reasonable speeds ($5\text{--}10 \text{ cm s}^{-1}$) but it is restricted to a very narrow current; this may be why export through the Fram Strait is only 0.03 Sv compared to the 0.09 Sv suggested by Aagaard and Carmack (1989). The Beaufort Gyre is too strong which may well be associated with the treatment of the surface stress in the very simple representation of ice dynamics. The Transpolar Drift has to go round the polar island. The simulation correctly shows westward motion around the Antarctic coast but, because of the lack of Coriolis turning in the

Fig. 8a, b As Fig. 7 but for the Southern Hemisphere sea ice

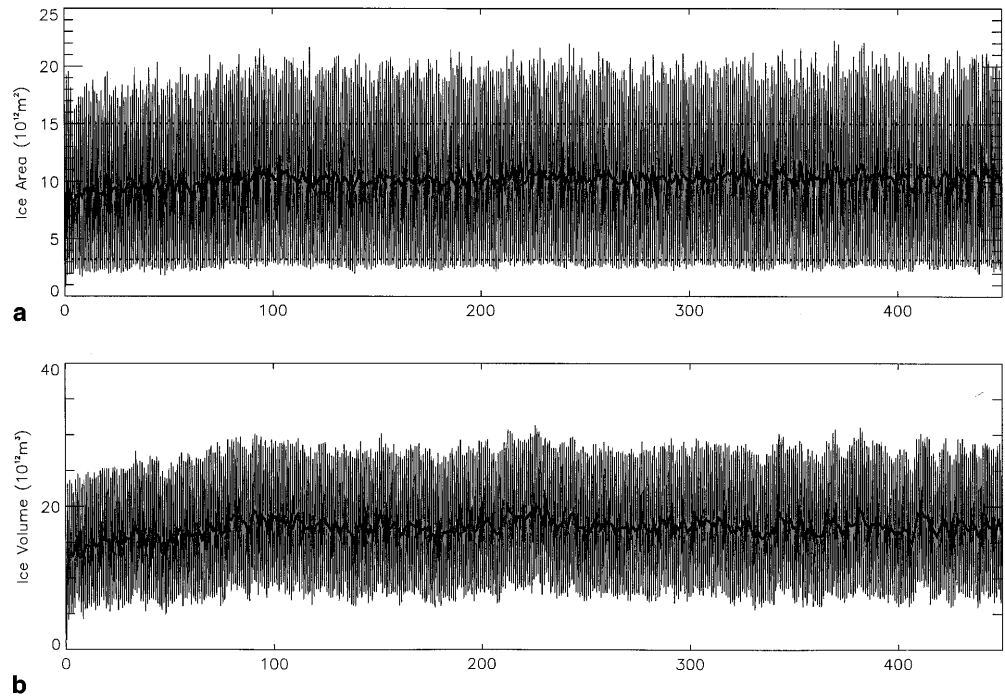
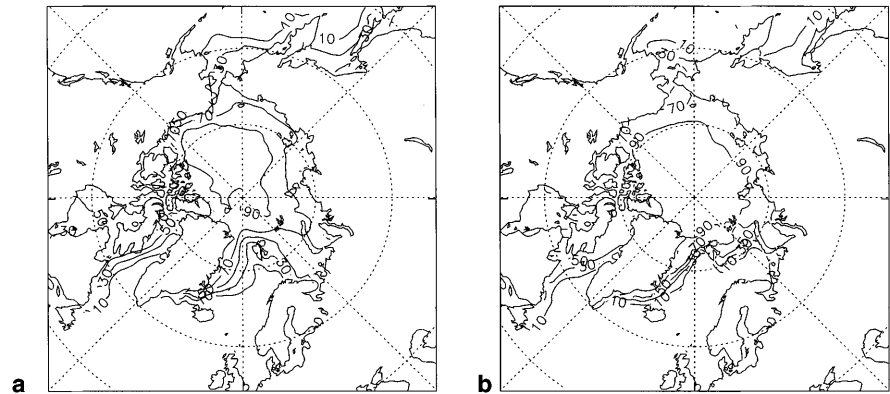


Fig. 9 **a** Modelled annual mean Northern Hemisphere sea ice concentrations for the years 361–400. Concentration is plotted as a percentage and the contour interval is 20%. **b** As **a** but from SSM/I data



simple ice model, the ice velocity tends to have an onshore component and hence divergence and ice production around the coast are too weak. Further simulations, with a revised free drift ice formulation in which the wind and water stress on freely moving ice are correctly represented, show an improved simulation of the ice velocity field in this region.

It is important to establish that, in addition to the good simulation of SST and sea ice extents, that the simulated surface heat fluxes and ocean heat transports are within the bounds set by the available observations. The following two sections therefore consider the evaluation of the momentum and heat budget in the coupled model.

6 Comparison of observed and modelled surface fluxes

In an atmospheric model, even when the SSTs are specified at their observed values, the calculation of the surface fluxes of momentum (i.e. wind stress), heat and fresh water depends upon a range of simulated model variables including low level winds, air temperatures and

humidities, as well as the cloud cover throughout the atmospheric column. These surface fluxes have been difficult to simulate realistically in climate and numerical weather prediction models (White 1996). It is these fluxes which constitute the forcing at the ocean surface in the coupled model and it is therefore important that they are accurately represented. The degree of accuracy required depends upon the sensitivity of the ocean simulation to the various forcings. Considerable work has been done to investigate this ocean sensitivity to wind stress forcing in the tropics (e.g. Harrison et al. 1989) but little has been done over the global oceans.

We consider the mean surface fluxes from a ‘stand alone’ integration of the atmospheric component of the coupled model for the 10 year period 1979 to 1988. The SSTs for this experiment were those used in the first Atmospheric Model Intercomparison Project (AMIP-Gates 1992) and in the remainder of the work this simulation will be referred to as ‘the AMIP simulation’. We

concentrate on the fluxes from this atmosphere simulation, rather than those of the coupled simulation, in order to help isolate systematic errors in the component models of the coupled system. Brief comparisons will also be made with coupled model surface fluxes.

This atmospheric model version is identical to that used in the coupled simulation apart from the specification of atmospheric trace gases (i.e. no additional 'tuning' was carried out). The AMIP simulation uses trace gas concentrations appropriate to the 1979 to 1988 period, whereas the coupled model uses pre-industrial values. The simulation of the atmospheric fields in the AMIP simulation is described in Pope et al. (1999).

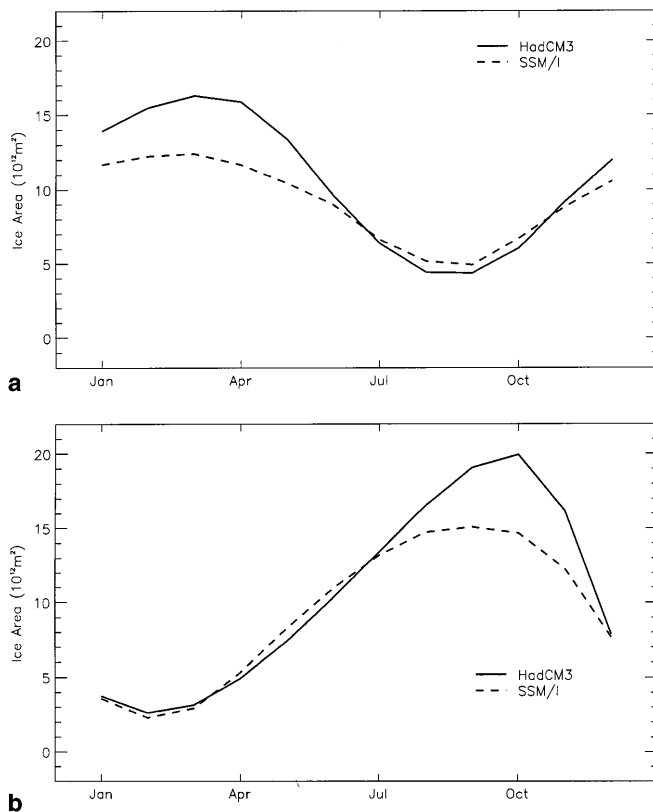
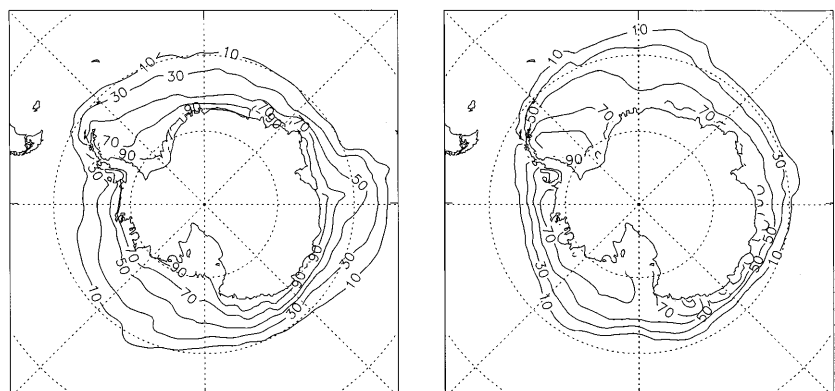


Fig. 10 **a** The mean monthly seasonal cycle of Northern Hemisphere sea ice area for the model years 361–400 and from SSM/I data. Units are 10^{12} m^2 . **b** As **a** but for the Southern Hemisphere

Fig. 11 **a** Modelled annual mean Southern Hemisphere sea ice concentrations for the years 361–400. Concentration is plotted as a percentage and the contour interval is 20%. **b** As **a** but from SSM/I data



Although our primary interest concerns the surface heat budget we will also attempt to evaluate the surface wind stress as this is clearly an important forcing function for the ocean. The freshwater budget will be evaluated in a separate paper. The surface wind stress fields from the model are compared with the climatologies of Hellerman and Rosenstein (HR) (1983) and Josey et al. (1996). The surface heat fluxes are compared with the climatological estimates of DS and Josey et al. (1996). The evaluation of the surface fluxes simulated by the atmospheric component of the coupled model is limited by the large uncertainties in the climatological estimates of these fluxes. Throughout the discussion that follows we will concentrate on estimates of the annual mean fluxes simply for clarity and brevity of presentation. These are also the fields that are of most relevance to the long term climate drift in the model. However, we note that the seasonal evolution of SST is also crucially dependent on the surface flux simulation (Gordon and Bottomley 1985).

6.1 The surface wind stress

The annual mean surface wind stress from the AMIP simulation is shown in Fig. 12a and the climatology of HR is shown for comparison in Fig. 12b. Overall the patterns of wind stress are well simulated. The trade wind regions have higher stresses in HR but this difference is probably within the uncertainty in the climatological estimates. Figure 12c shows the equivalent stress fields from the more recent estimates of Josey et al. (1996), and the model is in better agreement with this climatology in the trade wind regions of both hemispheres.

Major differences occur in the Southern Ocean where the model stresses are considerably larger than either climatology although very little data is available in this region. Comparison of low level model winds with equivalent fields from the ECMWF re-analysis climatology (ERA, Gibson et al. 1997) show a good agreement between the climate model and analysis fields in the Southern Ocean although, of course, the ERA will be largely unconstrained in this region and the analysis

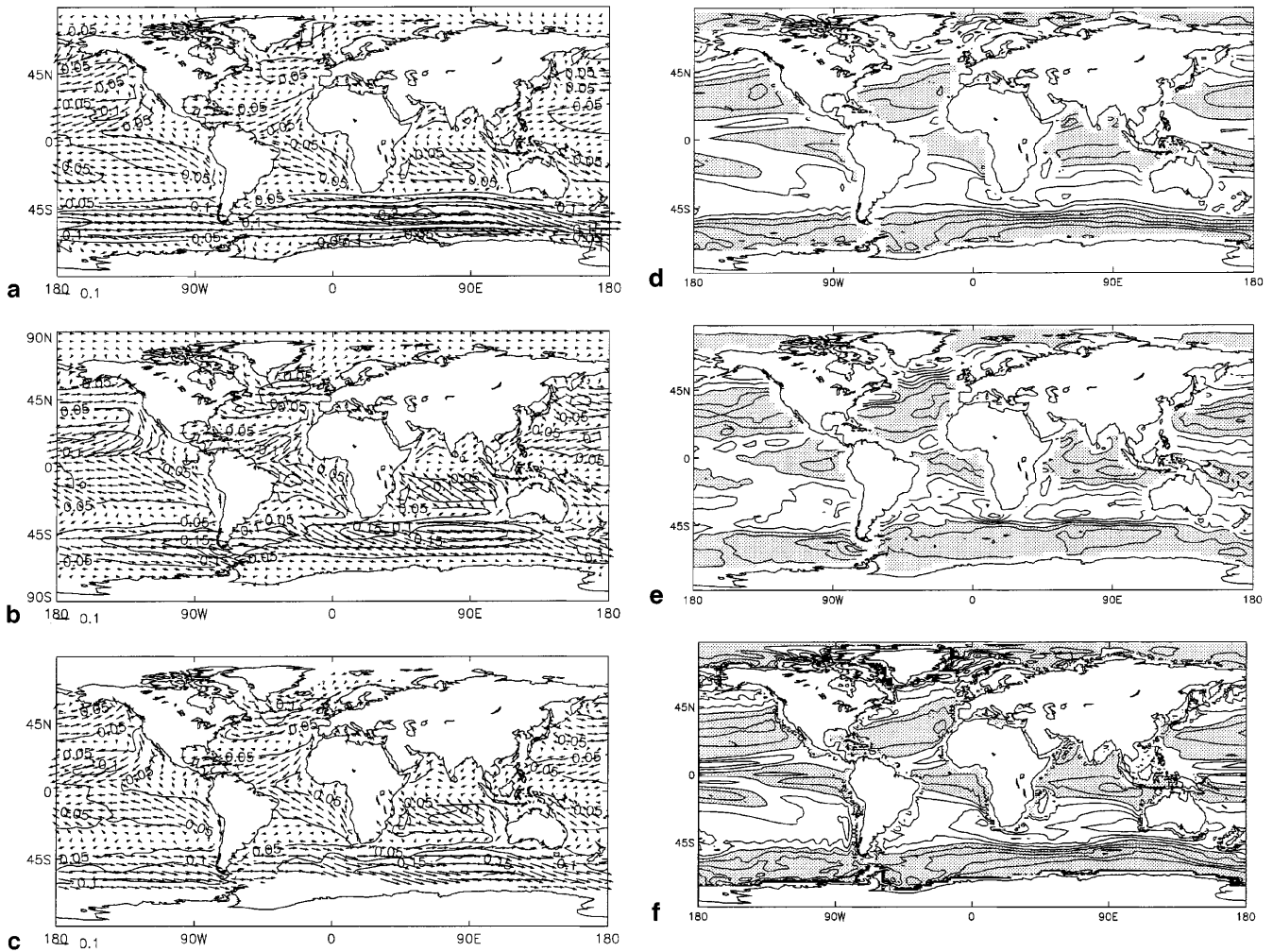


Fig. 12 **a** Ten year annual mean surface stress (Nm^{-2}) from the AMIP simulation with the HadAM3 model. **b** Annual mean surface stress from the Hellerman and Rosenstein (1983) climatology (Nm^{-2}). **c** Annual mean surface stress from the Josey et al. (1996) climatology (Nm^{-2}). **d** Ten year annual mean wind stress curl from the AMIP simulation with the HadAM3 model. (The contour interval is

$0.5 \times 10^{-7} \text{ Nm}^{-3}$ and negative regions are shaded.) **e** Annual mean wind stress curl from the Hellerman and Rosenstein (1983) climatology. (The contour interval is $0.5 \times 10^{-7} \text{ Nm}^{-3}$ and negative regions are shaded.) **f** Annual mean wind stress curl from years 81–120 of the HadCM3 coupled simulation (The contour interval is $0.5 \times 10^{-7} \text{ Nm}^{-3}$ and negative regions are shaded.)

may reflect systematic errors in the ECMWF model. Further data studies are required to be able properly to evaluate the model in this region.

There are also significant differences between the model and climatology in the North Atlantic where the simulated stresses in the westerlies are too weak. This is associated with a general high pressure bias at high latitudes, leading to too weak surface pressure gradients in mid-latitudes (see Pope et al. 1999 for details).

In the equatorial regions the Hellerman stresses are known to be too large (Josey et al. 1996). The model equatorial stresses appear to be too strong in the Pacific when compared with the ECMWF analysis and ship winds (not shown). This can, in part, explain the too cold SSTs in the modelled equatorial Pacific (Fig. 6).

The annual mean wind stress curl from the AMIP model and HR is shown in Fig. 12d, e. The overall pattern of the curl is well reproduced by the model and the positioning of the zero curl is represented realisti-

cally. In the north Pacific the zero windstress curl in part defines the separation point of the Kuroshio western boundary current (Hulbert et al. 1996). In reality, and in the coupled model (see Sect. 5), the Kuroshio extension is approximately coincident with the region of high horizontal SST gradient. The overall curl magnitudes are generally smaller in the model, compared to HR, which is consistent with the differences already noted in the stress magnitudes. The annual mean wind stress curl from the coupled model, meaned over the years 81–120, is shown in Fig. 12 f. The wind stress curl zero line has moved south in the western North Pacific when compared to the atmosphere only AMIP simulation described already with prescribed SSTs. This movement is consistent with the coupled model SST errors in the Kuroshio extension region of the North Pacific (Fig. 6a, b). Additional sensitivity experiments in which the ocean model was forced with HR climatological stresses suggest that this southward movement of the Kuroshio

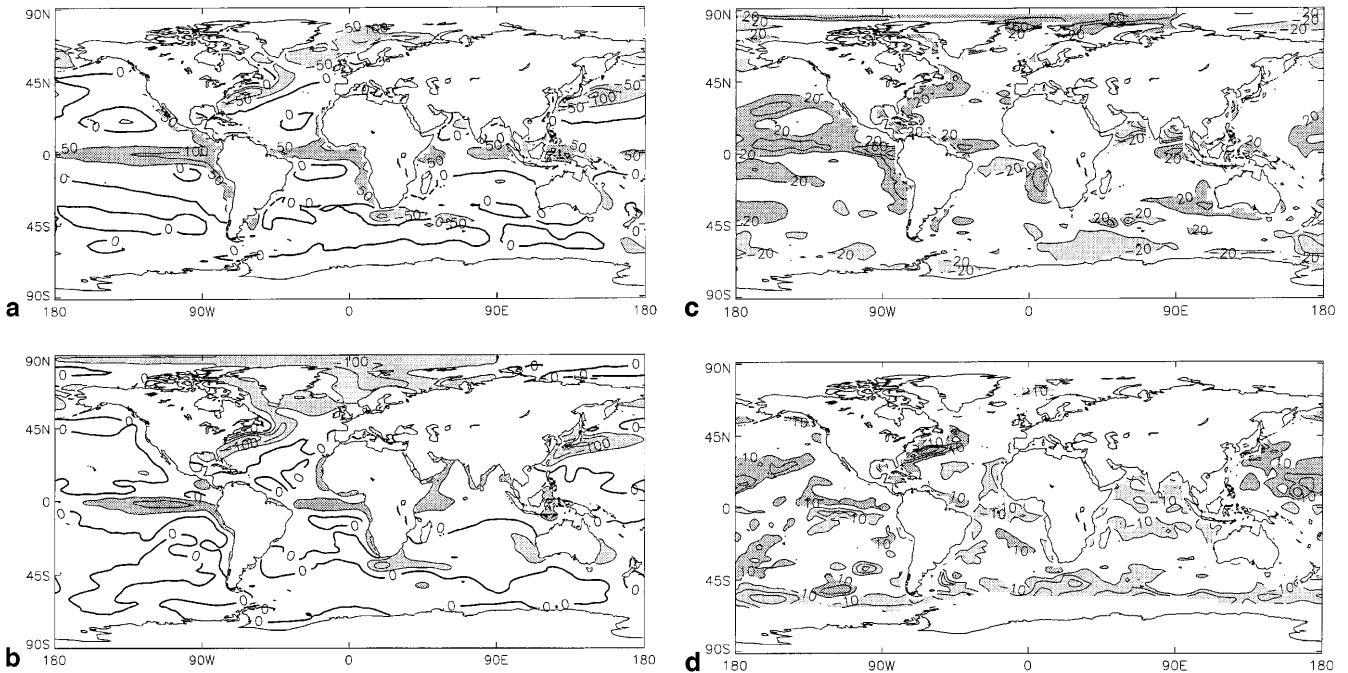


Fig. 13 **a** Ten year annual mean of the net surface heat flux from the AMIP simulation with the HadAM3 model (contour interval is 50 Wm^{-2}). **b** Annual mean surface heat flux from the Da Silva et al. (1994) climatology (contour interval is 50 Wm^{-2}). **c** Difference of

model minus Da Silva et al. (1994) annual mean net surface heat flux (contour interval is 20 Wm^{-2}). **d** Difference between the Josey et al. (1996) and Da Silva et al. (1994) annual mean uncorrected climatologies of net surface heat flux (contour interval is 10 Wm^{-2})

separation is a consequence of the model wind stress distribution. With the HR stresses the separation point was located further to the north and therefore closer to its observed position. In both the atmosphere only and coupled model there are considerably higher meridional gradients of wind stress curl in the Southern Ocean, compared to HR, and these are associated with the strong model stresses in this region.

6.2 The net surface heating

The net heat flux at the surface has four components; the net shortwave and longwave fluxes, the latent heat flux associated with surface evaporation, and the sensible heat flux. The shortwave and latent heat fluxes are the dominant contributions both in terms of magnitude and geographical variation. The net annual mean surface heating from the AMIP simulation and the corrected DS climatology are shown in Fig. 13a, b respectively. The corrections to the DS climatology are made to ensure a global balance of the surface heat budget (see Da Silva 1994 for details). A contour interval of 50 Wm^{-2} has been used in order to highlight the main features. The difference plot in Fig. 13c has a smaller contour interval (20 Wm^{-2}) to show greater spatial detail.

From Fig. 13a, b it can be seen that the major pattern features are well simulated by the model. In detail there are, however, some notable differences evident in the difference field in Fig. 13c. For comparison, Fig. 13d shows the difference between the Josey et al. (1996) un-

corrected total heat flux and the same quantity from the DS climatology. The uncorrected climatologies are used so that the differences in Fig. 13d are only due to the detailed calculation methods and not the correction procedures. There is a difference between the climatologies of around $10\text{--}20 \text{ Wm}^{-2}$ over much of the ocean which gives some measure of the level of uncertainty in the observational estimates. Note also, particularly in the Southern Ocean and the sub-tropical North Pacific, that the model is in better agreement with the SOC climatology.

The difference field in Fig. 13c shows that in the tropical east Pacific and Atlantic of the Southern Hemisphere the model has too much heat into the ocean surface. This is particularly marked on the Equator in the east Pacific, where the climatology shows the maximum heating some distance from the coast (Fig. 13b), whereas the model shows a large heating all the way to the coast (Fig. 13a). This feature is associated with the poor simulation of marine stratocumulus in these areas. A similar feature is also evident in the tropical Atlantic stratocumulus region. Figure 14a shows the difference in the net shortwave flux between the model and DS. Both the Atlantic and Pacific regions referred to stand out as having an excessive shortwave heat flux at the surface. Note that the Californian stratocumulus region also shows up clearly in the shortwave difference map. As shown earlier, when this atmospheric model is coupled to the ocean, these stratocumulus regions show SSTs that are too warm (see Fig. 6a, b).

Some other notable differences in Fig. 13c are the tropical and sub-tropical regions in the North Pacific

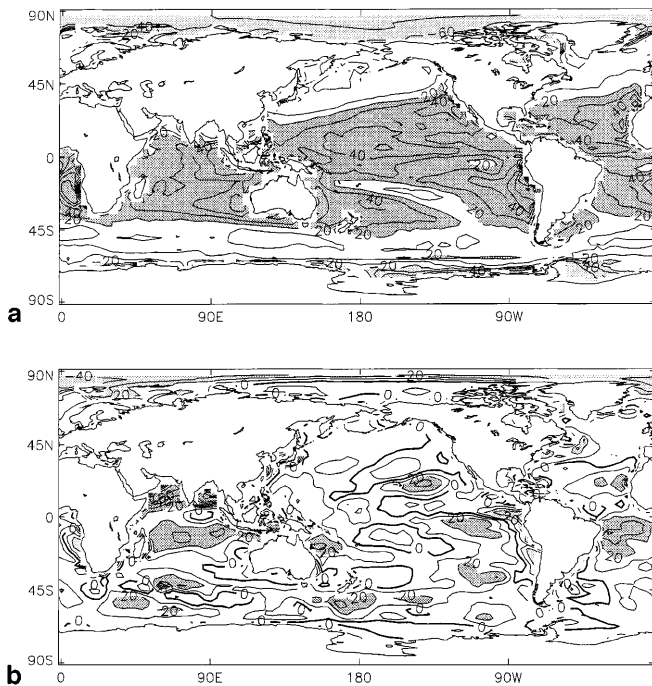


Fig. 14 **a** Difference between the ten year annual mean HadAM3 AMIP simulation of surface short wave flux and the Da Silva et al. (1994) climatology of the same quantity (the contour interval is 10 Wm^{-2} and values differences greater than 20 Wm^{-2} are shaded). **b** As **a** but for the surface latent heat flux

where there is more heat entering the ocean in the model than suggested by DS. Immediately under the peak of the trades (i.e. the region of maximum trade wind stress in Fig. 12, centred on 20°N , 135°W) this bias is reversed and the model exhibits a greater cooling than DS. In this region the overall increased heating over a broad area is a result of the larger surface short wave flux in the model (Fig. 14a), which is compensated for in the peak of the trades by enhanced surface evaporation (Fig. 14b). In the Indonesia/east Indian Ocean region Fig. 13c shows that the model gains too much heat, mainly as a consequence of the enhanced short wave flux, by more than 50 Wm^{-2} , in some areas.

The difference between the mean net surface heating in the coupled model for the years 361–400 and the AMIP simulation of the same quantity (Fig. 13a) is shown in Fig. 15. This figure shows the impact of the coupled model drift in SST on the net surface heat flux simulation. The region of strong equatorial Pacific heating extends further west in the coupled model and is more intense due to the excessive upwelling of cold water on the equator, which leads to a reduced surface evaporation. The excessive upwelling is consistent with too strong low level equatorial winds. Note also the large flux differences associated with the changed position of sharp horizontal SST gradient in the model. These regions are associated with the extension of the northern oceans' boundary currents and the ACC. In the North Pacific, where there is the largest SST error in the model, there is less heat loss in the coupled model than in the

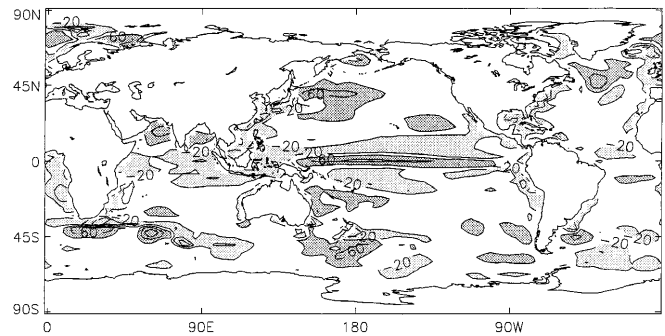


Fig. 15 Difference between the HadCM3 coupled model (year 361–400) and AMIP simulation of net surface annual mean heat flux (contour interval is 40 Wm^{-2} and differences greater than 20 Wm^{-2} are shaded)

AMIP simulation due to a reduction in surface evaporation as the modelled SST cools. Note also the greater cooling in the Pacific and Atlantic stratocumulus regions associated with the increased SST, and therefore surface latent heat loss, in the coupled model.

The large uncertainties in the climatological estimates need to be considered when assessing the comparisons made above. In particular, from a comparison of Fig. 14a, b, showing the shortwave and latent heat flux difference fields between the model and corrected DS, it is apparent that there is a net bias between the model and DS. The model shortwave fluxes are everywhere larger than those from DS, whereas the latent heat fluxes are in many regions larger in the model, leading to some cancellation. The net longwave heat losses (not shown) are also generally larger in the model than in corrected DS, globally by about 7 Wm^{-2} .

Both the climatological and modelled surface heat fluxes can easily be integrated to give the implied ocean heat transports. These transports are plotted in Fig. 16 (coupled model transports are also shown on the figure and will be discussed in the next section). In both cases for simplicity of comparison the heat transport at 65°N was assumed to be 0.1 PW (Da Silva et al. 1994). The un-corrected DS climatology has a global imbalance of 30 Wm^{-2} and the heat transports from the un-corrected fields are therefore meaningless. The DS heat transport curve plotted in Fig. 16 is from the corrected climatology which has a constraint of a zero global mean net heating imposed. The implied heat transport curve is therefore closed. The AMIP simulation, which used present day trace gas amounts and SSTs, has a net TOA imbalance of 3.5 Wm^{-2} , much of which will be reflected in the imbalance of the global mean surface heating (Pope et al. 1999) and the implied ocean heat transport is therefore not closed. For comparison with DS the model curve included in Fig. 16 uses the AMIP model transports that have been adjusted to ensure a zero global mean net surface heating.

There is considerable similarity between the corrected transports from the model and climatology. The largest differences occur in the Southern Hemisphere, in particular, in the sub-tropics. The role of the excessive heat

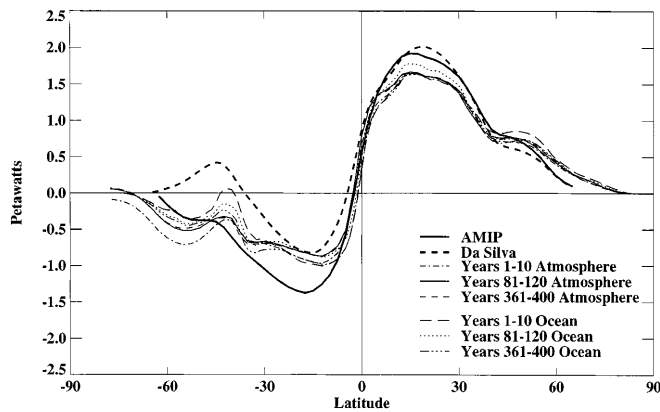


Fig. 16 The HadCM3 annual zonal mean poleward heat transport (PW) as a function of latitude at different times during the coupled simulation. The AMIP and Da Silva et al. (1994) values are also plotted. A correction was applied to the AMIP and Da Silva et al. (1994) fluxes to ensure a global mean balance when plotting the implied ocean heat transports. See text for details

input in the stratocumulus (Sc) regions in the AMIP simulation in determining the discrepancies in the Southern Hemisphere sub-tropical transport can be assessed by artificially correcting the heat input in these Sc regions back to DS. This calculation suggests that between the equator and 15°S an additional flux of approximately 0.4 PW of heat is entering the ocean in the AMIP simulation compared to DS. At 15°S this would account for 80% of the discrepancy in the implied ocean heat transport shown in Fig. 16. In part, the different sign of the heat transport in the Southern Ocean can be accounted for by the lower sub-tropical values. The correct slope of the transport curve between approximately 15°S and 40°S indicates an agreement between the model and DS fluxes. In addition, between 40°S and 60°S the net surface heating difference field between the model and DS (Fig. 13c) shows the model losing more heat to the atmosphere than the climatology, and this is reflected in the implied ocean heat transports. In fact, the DS climatology shows a net heat gain in this region. The observed fluxes in the Southern Ocean should be treated with considerable caution as the data coverage is very sparse.

7 Comparison of observed and modelled ocean heat transports

To complement the discussion of the atmospheric model surface heat budget, in this section we consider the ability of the ocean model to reproduce the geographical distribution of global ocean heat transports when compared with direct ocean observations.

7.1 Global distribution of poleward heat transports

In the AMIP simulation present day SSTs and atmospheric greenhouse gas concentrations were used. In the coupled model pre-industrial gas concentrations are

employed which reduce the TOA flux imbalance by approximately $1.5\text{--}2.0\text{ Wm}^{-2}$. Therefore the global mean imbalance at the start of the coupled simulation is approximately half the imbalance in the AMIP experiment. As the coupled simulation proceeds, the TOA imbalance quickly reduces to less than 0.5 Wm^{-2} (Fig. 2). This happens due to the rapid adjustment of the SSTs in the coupled model.

Shown on Fig. 16 are a number of estimates of model poleward ocean heat transport using net fluxes from the atmosphere component of the coupled model from near the beginning of the coupled HadCM3 simulation, and after 100 and 400 years. Note also that in Fig. 16 the assumption of 0.1 PW heat transport at 65°N has not been made in plotting the curves from the coupled model, in fact, the model suggests a transport of approximately 0.25 PW at this latitude. For comparison the ocean heat transports as calculated within the ocean component of the coupled model are also included in Fig. 16 for the same time periods. As noted earlier in Sect. 2, the long term mean estimates of heat transports from the surface fluxes and from the ocean model must agree as the model reaches equilibrium. A critical issue is whether this state can be achieved whilst maintaining a realistic climate simulation in other respects.

Although the actual and implied heat transports differ by up to 0.2 PW in the sub-tropics, comparison of Fig. 16 with the equivalent plot for the earlier flux adjusted versions (HadCM2 and UKTR) in Fig. 1, shows that there is a much greater consistency in the new model between the atmospheric model surface fluxes and the ocean heat transports.

The modelled mean global heat transports for the years 361–400 are shown again in Fig. 17 and are broken down into component parts due to the zonal mean meridional overturning circulation, the gyre circulation (defined as the deviation from the zonal mean), the along isopycnal diffusive component and the eddy transports due to the Gent and McWilliams (1990) scheme.

The meridional overturning flux contains the contribution from the near surface Ekman transport, and this dominates in the tropics. In the model the contribution from the along isopycnal diffusive term and from the bolus velocity advection is small at most latitudes, and only makes a significant contribution to the zonal mean in the region of the Deacon Cell in the Southern Ocean. The smallness of the bolus advection term should not be interpreted as meaning the Gent and McWilliams (1990) parametrisation does not influence the heat transports. What it shows, at least in this model, is that the most important aspect of the Gent and McWilliams (1990) parametrisation is its ability to better maintain water mass properties (which comes about because of the absence of any background horizontal diffusion in the scheme), rather than it producing a significant heat flux directly due to eddies. The along isopycnal diffusive flux also makes a small contribution to the depth averaged meridional transport.

7.2 Comparison with ocean observations

To evaluate the modelled ocean heat transports the model results are compared with those obtained from direct ocean observations. Figure 18 shows recent direct estimates of meridional ocean heat transports from observations across the marked latitudes, with the authors indicated by their initials. The stars indicate that the volume flux across the section is non-zero and the numbers indicate transports of temperature normalised to PW. The estimates are calculated from the instantaneous profiles of velocity and temperature measured on a single hydrographic section. For comparison, Fig. 19a, b shows the advective heat transport across the marked latitudes for the mean of years 81–120 and 361–400 from the coupled model. In addition to the observed sections shown in Fig. 18, some extra model sections have been included in Fig. 19a, b. The model estimates are calculated from the mean velocity and temperature profiles

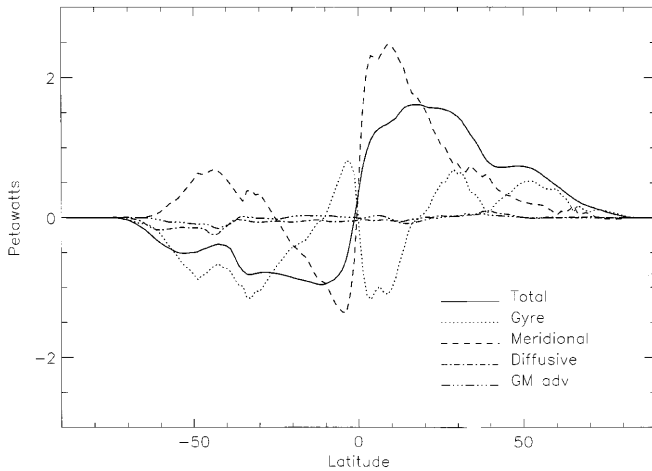


Fig. 17 HadCM3 component contributions of the annual zonal mean poleward heat transport (years 361–400) as a function of latitude (PW). Component contributions from the gyre, meridional overturning, along isopycnal diffusion and eddy bolus velocities are shown

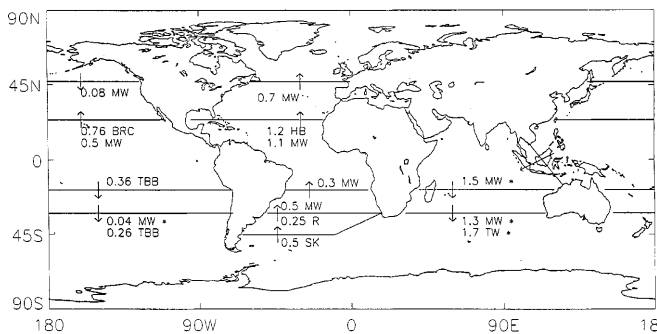


Fig. 18 Direct ocean observational estimates of the meridional heat transport at marked latitudes. The authors for particular sections are indicated by initials; *MW*, Macdonald and Wunsch (1996), *BRC*, Bryden et al. (1991), *HB*, Hall and Bryden (1982), *TBB*, Tsimplis et al. (1996), *R*, Rintoul (1991), *TW*, Toole and Warren (1993) and *SK*, Saunders and King (1995)

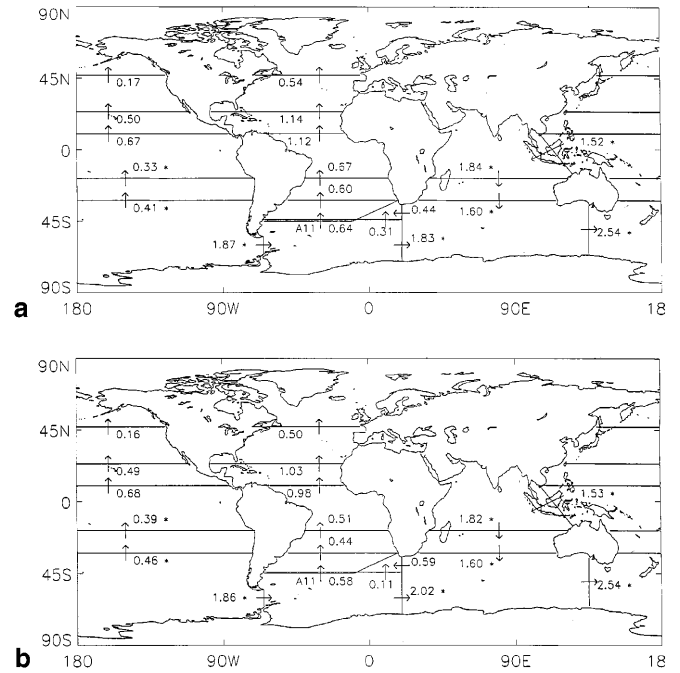


Fig. 19 **a** As Fig. 18 but for the HadCM3 coupled model for years 81–120. **b** As Fig. 18 but for the HadCM3 coupled model for years 361–400. In both **a** and **b** some additional model sections are included

for the period indicated. Observational estimates make the assumption that the velocity and temperature profiles measured on a single hydrographic section are representative of the long-term mean velocity and temperature profiles. The eddy heat flux cannot be estimated accurately from observations. Here we consider only the large-scale advective heat transport.

A comparison of the heat transport for the two time periods (years 81–120 and 361–400) shows there to be a reduction in the Atlantic heat transport by $O(0.1 \text{ PW})$ in the later period, while transports in the other basins are generally unchanged. Comparing Figs. 18 and 19a suggests that in the North Atlantic and North Pacific the ocean heat transports produced by the coupled model are in reasonable agreement with the observational estimates. In the South Atlantic the model shows less divergence in the heat transport than the observational estimates suggest, and the magnitude of the northward heat transport near 30°S is larger than all the published estimates with the exception of that by Fu (1981) of 0.8 PW (not shown). For a fuller discussion of the model heat transports in the South Atlantic see Banks (To appear 1999).

In the Indian and South Pacific oceans the Indonesian through-flow (ITP) gives a non-zero mass transport across the sections and the numbers indicated on Figs. 18 and 19 across these oceans are temperature transports (multiplied by $\rho_0 C_p$). (Note the transport $\rho_0 C_p \int_{\text{section}} vT dx dz$ is not invariant under change of temperature scale when $\rho_0 \int_{\text{section}} v dx dz \neq 0$, the mass transport across the section is non-zero.)

In the Indian Ocean the temperature transport is southwards as seen in the observations but is larger in magnitude, while in the South Pacific the temperature transport is in the opposite direction to most of the observations. These differences are associated with the magnitude of the ITF in the model, 24 Sv, which is larger than the observational range of 0–20 Sv (Wijffels et al. 1996), and dominates the temperature transports in the basins. The combined Indo-Pacific heat transports are in reasonable agreement with the observations.

7.3 Comparison with the Atlantic and Pacific 24°N sections

To further validate the model a more detailed comparison of the components contributing to the total heat transports has been carried out for the sections at 24°N in the Atlantic and the Pacific. These 24°N sections are considered to provide robust observational estimates of the ocean heat transport because they are located in the centre of the sub-tropical gyre and the eddy heat transport is considered small compared to the total heat transport. Table 1 gives the breakdown of the total heat transport into its western boundary component (WBC), Ekman (wind driven) and interior components. In the model the WBC is defined by maximising the volume transport from the western boundary.

Comparing the components from HadCM3 at 24°N with the observational estimates of Hall and Bryden (1982) for the Atlantic, and Bryden et al. (1991) for the Pacific, we can see similar differences for both sections; an overestimate of the heat transported northward by the WBC and of the heat transported southward by the interior and an underestimate of the heat transported northward by the Ekman layer. The underestimate of the Ekman layer heat transport is related to the reduced Ekman transport across the section; 3.8 Sv (Atlantic)

and 7.5 Sv (Pacific), compared with 5.0 Sv (Atlantic) and 12.0 Sv (Pacific) in the observational estimates. The observational estimates are based on the climatological windstresses from Hellerman and Rosenstein (1983) which may bias the Ekman transport to larger values. For example, across 24°N in the Pacific, calculating the Ekman transport from the Josey et al. (1996) climatology rather than HR leads to a reduction of 3 Sv. The remaining 1.5 Sv discrepancy between model and observations is relatively small.

The overturning component of the heat transport in HadCM3 is 1.07 PW for the Atlantic, accounting for 93% of the total heat transport (compared with the observational estimate of 1.28 PW accounting for 105% of the total). In the Pacific, the overturning component is 0.14 PW accounting for 28% of the total (compared with the observational estimate of 0.38 PW which accounts for 50% of the total). In both cases, not only is the magnitude of the overturning component smaller than the observed estimate but it suggests a greater role for the gyre component in the model than is indicated by the observations.

Boning et al. (1996) compared the strength of the North Atlantic Deep Water (NADW) cell at 25°N with the poleward heat transport at the same latitude, from a variety of models with different resolutions and forcings. Although there is a considerable range amongst the models in the value of the heat transport at 25°N, there is a simple linear relationship between the strength of the NADW cell and the poleward heat transport that fits all the model results. The overturning meridional/depth stream function for the North Atlantic in the coupled model for the periods years 81–120 and 361–400, is shown in Fig. 20a, b. The strength of the NADW cell is stable, whereas a deeper cell clearly ‘spins up’ over the 400 years of the integration. The strength of the NADW cell at 24°N is between 16–18 Sv, with a corresponding heat transport of 1.1 PW. These values also fit the simple relationship between the overturning and heat transport derived in Boning et al. (1996). Hall and Bryden (1982) estimate the magnitude of the NADW cell to be 19 Sv (integrating their Table 5). The model transport of NADW agrees well with observations at three key locations where there are robust observational estimates (Wood et al. 1999).

The volume transports at 24°N in the Atlantic can also be compared with observations for different temperature classes. Table 2 shows the transports in temperature classes for the model during the years 81–120 and 361–400 compared with the observational estimate of Hall and Bryden (1982).

Compared with Hall and Bryden (1982), HadCM3 transports more water northwards in the warmest temperature class and less in the two classes below this. By the later period, about 2 Sv of warmest water has moved to the cooler temperature classes.

For the earlier period, there is some indication of southward flow of NADW moving to the temperature class $4 < \theta < 7$ °C. This trend appears to continue, so

Table 1 Values for the heat transport components (PW) for the 40 year mean of years 81–120 from HadCM3 and the range (based on the decadal means for years 81–120). The total heat transport is split into components in two different ways: (a) Western Boundary Current (WBC)/Ekman/Interior and (b) Overturning/Gyre

	Observed	40 y mean	Range
Atlantic 24°N			
WBC	1.73	2.00	1.95–2.02
Ekman	0.42	0.28	0.27–0.29
Interior	–0.93	–1.14	–1.15––1.10
Total	1.22	1.14	1.12–1.16
Overturning	1.28	1.07	1.04–1.09
Gyre	–0.06	0.07	0.08–0.07
Pacific 24°N			
WBC	1.73	2.32	2.23–2.37
Ekman	0.93	0.67	0.66–0.68
Interior	–1.91	–2.49	–2.41––2.55
Total	0.76	0.50	0.49–0.51
Overturning	0.38	0.14	0.13–0.16
Gyre	0.38	0.36	0.36–0.35

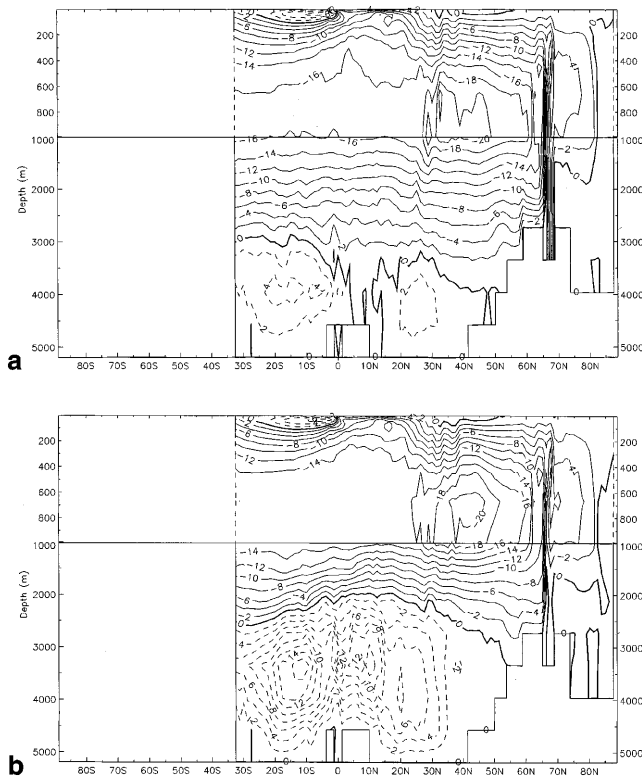


Fig. 20 **a** Annual mean North Atlantic overturning stream function (Sv) for years 81–120 of the HadCM3 coupled model. **b** As **a** but for years 361–400

Table 2 Northward volume transport in temperature classes (Sv)

	Hall and Bryden Observed	Model years 81–120	Model years 361–400
$\theta > 17^{\circ}\text{C}$	8.9	13.8	11.4
$12 < \theta < 17^{\circ}\text{C}$	2.5	0.6	0.9
$7 < \theta < 12^{\circ}\text{C}$	6.6	1.3	3.1
$4 < \theta < 7^{\circ}\text{C}$	-2.3	-6.8	-16.4
$\theta < 4^{\circ}\text{C}$	-15.6	-8.8	2.0

that during the later period all NADW transport is in the warmer class. This is consistent with the reduction in northward heat transport between the two periods and with the shallower depth of the NADW cell in the later period. Overall, the adjustment of the transport in temperature classes is consistent with a reduction in the effective temperature difference of the overturning. The detailed vertical structure of the temperature drift in the coupled model is complex and has yet to be analysed in detail. The global drifts shown in Fig. 3a show the near surface cooling, with a larger warming below this that extends down to 1000 m. The equivalent diagnostic for the North Atlantic (not shown), shows a similar cooling at the surface (see Fig. 6b) but with an even larger subsurface warming, peaking at 2°C around 1000 m. The Atlantic warming extends to all deeper levels.

In summary, the overturning cell in the Atlantic is slightly too weak in HadCM3 when compared to Hall and Bryden (1982). This is consistent with the difference in total heat transport of $O(0.1 \text{ PW})$ between model and observed as predicted by the simple linear relationship in Boning et al. (1996). The larger magnitude in the observations may be partly attributable to the Ekman transport calculated from HR which is larger than the model Ekman transport. Macdonald (1995) found that reducing the Ekman transport from 5.4 Sv (from HR stresses) to 4.2 Sv (ECMWF stresses) reduced the total heat transport by 0.13 PW. Finally, the percentage of the heat transport explained by the overturning and gyre components is probably related to the structure of the western boundary current in the model. With a horizontal resolution of 1.25° the boundary currents cannot be simulated realistically.

8 Summary

The HadCM3 Hadley Centre coupled model has a much improved SST and sea ice climatology when compared to earlier Hadley Centre models without flux adjustments (Gregory and Mitchell 1997). One of the key features in this advance has been the improved simulation of both the surface heat fluxes and the ocean poleward heat transports, which in the HadCM3 model are in broad agreement with the observed estimates. This was not the case in earlier versions of the model. In HadCM3 the heat transports needed by the atmospheric component to maintain its present day climate agree reasonably well with the heat transports simulated by the ocean model. As suggested by Weaver and Hughes (1996) this leads to a stable model. Boville and Gent (1998) and Bryan (1998) also conclude that this consistency of fluxes is, in part, responsible for the stability of their 300 year simulation with the non-flux adjusted CSM1 model. They also attribute the stability to the initialisation procedure used. In HadCM3 we have no pre-coupled initialisation and this is not a pre-requisite for a stable simulation with this model.

We conclude the following:

a. The atmospheric model, run in atmosphere-only mode, is capable of a realistic simulation of the surface heat flux and surface stress fields. This in itself is a significant achievement in atmospheric model performance. One major limitation here is the quality of the flux climatologies used to evaluate the model.

b. When coupled, the ocean model approximately maintains the poleward heat transports as implied by the atmosphere only case. That is, the system is self-consistent and there is no need for the SST to undergo large drifts in order to rebalance the heat budget. Heat flux adjustments are therefore not required and the coupled model can be initialised directly from an observed ocean state without the need for separate equilibration of the ocean model. Although there are salinity drifts in the model these are not large enough to significantly de-

riorate the ocean circulation and thereby the atmosphere climate simulation.

c. The ocean component of the coupled model simulates poleward heat transports in broad agreement with estimates from oceanographic sections.

The SST pattern in the model stabilises after only a few decades. The sea ice simulation is encouraging and the improvement in ocean heat transports in the model has an important consequence in improving the position of the ice edge. The strength of the thermohaline circulation in the model is stable although there is a drift in its vertical structure through the integration. There is also significant multi-decadal variability, with an amplitude of 1–2 Sv in the maximum North Atlantic overturning streamfunction (Wood et al. 1999).

A major emphasis of this study has been the evaluation of the surface heat exchange and ocean heat transport processes in the coupled model. It is important in coupled models that this kind of analysis is carried out to ensure that the mechanisms in the model, at least those in determining the mean balanced state, are realistic. This is necessary if we are to have confidence in numerical experiments to understand climate and predict climate change using such models.

Appendix A

The ocean model used in HadCM3 has a number of sub-gridscale parametrisations and details of these are given in this appendix.

A1 Vertical mixing of momentum

Momentum is mixed vertically using a K-Theory flux-gradient parametrisation. In the upper ocean mixed layer a simplified version of the Large et al. (1994) non-local parametrisation is used. Based on results from large eddy simulations, K_m is simply represented as a quadratic function of the depth z (see Large et al. 1994 for details). A boundary layer scaling depth h is determined as the depth where the gradient Richardson number Ri exceeds a critical value of 0.3, and the diffusion coefficient K_m is specified by:

$$K_m(z/h) = a_0 + a_1(z/h) + a_2(z/h)^2.$$

The coefficients a_0 , a_1 and a_2 are determined by the requirement that near to the surface as $z \rightarrow 0$, $K_m(z) \approx ku^*z$ (k is von Karman's constant, u^* is the ocean friction velocity) and that K_m is continuous at the depth h . To prevent large K_m values in deep convective layers a maximum value of 80 m is imposed on h . The use of this formulation leads to a realistic simulation of the surface current direction. This is particularly important in areas of sea ice since in the model the direction of the ice drift is determined by the ocean currents.

In the region $z > h$, below the surface layer, the Pacanowski and Philander (1981) K-Theory parametrisation is used. The mixing coefficient K_m is the sum of a term dependent on the local gradient Richardson number and a depth independent background value of $1.0 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$. The Richardson number dependent part only makes significant contributions in the equatorial zone because, at 1.25° resolution, in all other regions current shears below the mixed layer are very small. (Note. In the earlier HadCM2 version of this model, Johns et al. 1997, the Pacanowski and Philander (1981) scheme was also used to mix momentum in the surface boundary layer. In the upper layer it led to small mixing coefficients and implied Ekman depths of less than the top model level at 10 m. Surface current directions were therefore poorly simulated. This

has been improved in HadCM3 by the inclusion of the simplified Large et al. 1994 scheme described.)

A2 Vertical mixing of tracers

The mixing is carried out by the K-Theory diffusion described using the simplified Large et al. (1994) scheme in the mixed layer (see later) and below this the Pacanowski and Philander (1981) scheme. The background tracer diffusion coefficient is now a function of depth. This depth dependent diffusivity is the same as that used in HadCM2 and is shown in Table A for completeness. This profile is a close fit to the theoretical/observational estimate of Kraus (1990), and the near surface values are in good agreement with recent direct measurements in the upper 1000 m of the ocean (Ledwell et al. 1993).

In the upper ocean boundary layer a hybrid tracer mixing scheme is used. In order to better simulate the upper ocean mixed layer structure, in addition to the diffusive mixing, a Kraus-Turner (1967) energetics parametrisation is also used (Gordon and Bottomley 1985). The input of energy from the wind that is available for mixing is parameterised by $\lambda \rho_0 u^*$ (the wind mixing energy), ρ_0 is a reference sea water density and λ a dimensionless constant of order unity. For a water column of uniform density, the turbulent energy available for mixing at a depth z is also decayed exponentially via $\exp(-z/\delta)$. This is included to limit the effects of surface induced mixing as the mixed layer deepens (see Kraus 1977 for discussion). The constants λ and δ are determined to give the same dissipation as that observed during the MILE study (Davis et al. 1985).

This turbulent energy is then used to mix the vertical temperature and salinity profiles so that the work done against gravity in mixing down lighter water balances the energy input. The partial mixing scheme of Thompson (1976) is used rather than carrying an explicit mixed layer depth. A simple convective adjustment is implemented when there is a negative surface buoyancy forcing. The convection is partially penetrative in that 15% of the energy released by convection is used in further deepening (Kraus 1977). In the layer which is well mixed by the KT scheme the diffusive mixing of tracers is trivial as the water column is already fully mixed.

A3 Eddy mixing

For eddies to be resolved explicitly would require a resolution much finer than that currently possible in climate models. The eddy

Table A Depth and thickness of model levels and the tracer diffusivities. See text for explanation

Level	Depth (m)	Thickness (m)	Vertical diffusivity ($10^{-5} \text{ m}^2 \text{ s}^{-1}$)
1	5.0	10.0	–
2	15.0	10.0	1.03
3	25.0	10.0	1.06
4	35.1	10.2	1.08
5	47.9	15.3	1.11
6	67.0	23.0	1.16
7	95.8	34.5	1.22
8	138.9	51.8	1.32
9	203.7	77.8	1.46
10	301.0	116.8	1.68
11	447.1	175.3	2.01
12	666.3	263.2	2.50
13	995.6	395.3	3.23
14	1501.0	615.0	4.34
15	2116.0	615.0	6.06
16	2731.0	615.0	7.79
17	3347.0	615.0	9.51
18	3962.0	615.0	11.23
19	4577.0	616.0	12.95
20	5193.0	616.0	14.68

effects must therefore be parameterised. The Gent and McWilliams (1990) (GM) scheme parameterises sub-grid-scale eddy activity by removing potential energy in an adiabatic manner by diffusion of isopycnal layer thickness. Earlier experiments with the 1.25° ocean model showed that it is not appropriate to set a globally constant thickness diffusion coefficient. This was because in order to maintain both the sharp horizontal density gradients associated with the North Atlantic Current and Kuroshio extension and, at the same time, simulate significant eddy fluxes in the Antarctic Circumpolar Current (ACC), different thickness diffusion coefficients are required (see Wright 1997 for details). In the model described here, the formulation of Visbeck et al. (1997) is therefore used in which the thickness diffusion coefficient is determined locally.

Visbeck et al. (1997) highlight three cases where different values of κ are appropriate. They suggest values of $2000 \text{ m}^2 \text{ s}^{-1}$ in a wind driven channel (analogous to the ACC), $500 \text{ m}^2 \text{ s}^{-1}$ in a frontal zone (analogous to the boundary currents) and $300 \text{ m}^2 \text{ s}^{-1}$ in a convective chimney. These values are in broad agreement with those required by the 1.25° ocean model to maintain the climatological temperature and salinity structure of the ACC and western boundary currents (Wright 1997).

Visbeck et al. (1997) suggest a time scale for the growth of baroclinic instability of $T^{-2} = f\bar{Ri}^{-1}$ where T is the time scale for the growth of eddies, f is the Coriolis parameter and $\bar{Ri}^{-1}$ is the inverse depth averaged Richardson number. Following Treguier et al. (1997), the vertical average is taken over the upper ocean between 100 m and 2000 m. Following Visbeck et al. (1997) the eddy transfer coefficient is then defined as $\kappa = \alpha T^{-1} l^2$ where α is a constant equal to 0.015 and l is a length scale chosen to be the width of the baroclinic zone. Following Visbeck et al. (1997) this length scale is determined from the horizontal distribution of the inverse time scale T^{-1} . A baroclinic zone was defined as a region where the inverse timescale is greater than $1.4 \times 10^{-6} \text{ s}^{-1}$, this number being determined by experiment. A minimum κ of $350 \text{ m}^2 \text{ s}^{-1}$ and a maximum of $2000 \text{ m}^2 \text{ s}^{-1}$ are allowed.

The thickness diffusivity values simulated by the model (Wright 1997) are in general agreement, both in magnitude and position, with those suggested by both Visbeck et al. (1997) and Treguier et al. (1997). With the spatially and time varying thickness diffusion coefficient, the GM scheme enhances mixing in regions of baroclinic instability. This also allows the model to maintain the tight density gradients in the northern gyres, whilst simulating a significant eddy flux across the ACC.

The Griffies et al. (1998) numerical implementation of the Redi (1982) isopycnal mixing scheme is also employed, as this leads to a considerable reduction in grid point noise compared to the original Cox (1987) implementation of the Redi (1982) scheme. The along isopycnal diffusion coefficient is $1000 \text{ m}^2 \text{ s}^{-1}$ and does not vary with depth. There is no explicit horizontal tracer diffusion in the model. The coefficient of horizontal momentum viscosity is formulated as $A_M = A_1 + A_2 \cos(\text{latitude})$ with $A_1 = A_2 = 3000 \text{ m}^2 \text{ s}^{-1}$. The latitude dependence of A_M is included to prevent numerical instability at high latitudes as the grid lines converge, while still resolving mid-latitude western boundary currents.

A4 The simple overflow scheme

One of the major systematic errors in an earlier version of the Hadley Centre $1.25^\circ \times 1.25^\circ$ global ocean model was a too-zonal simulation of the North Atlantic Current (NAC) (Wright and Gordon 1997). In the coupled model this would clearly have implications for the SST and therefore for the simulation of climate over Western Europe.

A similar problem was observed in the comparison of a one degree North Atlantic version of the Hadley Centre model with an equivalent version of the MICOM isopycnal model (Roberts et al. 1996). In the Hadley Centre model the overflow water across the Greenland–Iceland–Scotland ridge tended to mix with the water to the south of the sills. There were two sources of this erroneous mixing. The first was caused by a numerical restriction in the

isopycnal mixing scheme. In the Cox (1987) implementation of the scheme used in this earlier model version, a maximum value was imposed on the isopycnal slope used in calculating the diffusivity. The imposition of this maximum slope can lead to a large diapycnal component to the diffusion in regions of large horizontal changes in density (such as the overflow regions).

The second problem occurred due to the parametrisation of convective overturning in the model. In the convective mixing scheme, if two boxes were unstable then they were simply mixed together and so on down the column to end up with a stable profile. This led to any dense water entering near the top of the column being mixed out with the underlying water instead of running down the slope as a boundary current.

In order to test the hypothesis that the southward shift of the NAC was associated with the wrong water mass properties of the overflow water to the south of the sills, a number of sensitivity studies were carried out. These concentrated on the imposition of buoyancy anomalies in the sub-polar gyre, essentially to force the water to sink. The main result from these sensitivity experiments was to show that if the overflow water can be caused to sink to the right depths then the NAC problem is alleviated (Wright and Gordon 1997).

Two schemes have been developed in recent years to alleviate some of the erroneous overflow mixing problems described. The first is the Gerdes et al. (1991) (hereafter GKW) modification to the isopycnal mixing used in the model. In the GKW formulation the restriction of a maximum slope is relaxed, but the diffusivity along the isopycnals is reduced as the slope is increased. This avoids the generation of a diapycnal mixing component to the diffusion which occurs in the standard version of the scheme.

The scheme of Roether et al. (1994) aims to address the other aspect of the erroneous mixing in the model. As noted above the standard convective mixing leads to any dense water entering near the top of the column being mixed out with the underlying water. The Roether et al. (1994) scheme instead finds the level of neutral buoyancy for the overflow water and mixes the water into this model level. All the intervening water is then moved up in the water column to replace the dense water which has been mixed in. This way there is less mixing with intervening levels. This scheme was implemented only in the overflow region (approximately $0\text{--}45^\circ\text{W}$ and $58\text{--}70^\circ\text{N}$). To avoid excessive mixing as the overflow water ‘steps down’ the model bottom slope, the topography to the south of the overflow sills was modified so that the drop on the southern side of the sills takes place over a single gridpoint.

A 10 year simulation with the ocean only model showed that the inclusion of these schemes led to a greater sinking of the overflow water. However, the results were also very sensitive to the surface forcing used in the model. A short 20 years coupled simulation in which the overflow parametrisation was removed was run parallel to the HadCM3 simulation described in this paper. After 10 years the inclusion of the overflow scheme leads to a greater volume of dense bottom water in the sub-polar gyre and the vertical density gradients in the upper 1000 m are better maintained. In the simulation without the overflow scheme the overflow water mixes with the upper layers and diffuses the density gradient. After 20 years it is difficult to assess the impact of the scheme on watermasses in the sub-polar gyre because of the high level of natural variability in the coupled model; (e.g. after 20 years the maximum Atlantic overturning streamfunction changes by less than 0.5 Sv between the simulations with and without the overflow scheme and this is well within the decadal variability of the model). A longer century time scale simulation and significance tests would be necessary to definitively assess the impact of this and the other parametrisations used in the coupled model.

A5 The Mediterranean outflow scheme

The Mediterranean outflow parametrisation attempts to simulate the exchange of water between the Mediterranean and the Atlantic. At each level between 1 and 13 in the model (approximately the upper 1000 m), an average is formed of tracer values for two points

in the Mediterranean and two in the Atlantic adjacent to the Straits of Gibraltar (denoted by $\bar{T}$ below). For the temperature (T_j) at each of the four points a rate of change is calculated as:

$$\frac{\partial \bar{T}_j}{\partial t} = \mu(T_j - \bar{T}).$$

The constant μ reflects how much of the gridbox should be mixed in a time step. Hence, near the surface, Mediterranean points become fresher due to the fresher Atlantic water, and at depth Atlantic points become more saline. The constant μ is chosen to simulate a transport of 1 Sv through the Straits of Gibraltar.

Appendix B: data transfer between grids

Particular problems in the techniques for exchanging information between the ocean and the atmosphere arise from the models being on different horizontal grids. The information passed from the ocean to the atmosphere describes the state of the surface (temperature and sea ice parameters). Because the sea ice fields can have sharp discontinuities, bilinear interpolation from the higher-resolution ocean grid is not considered satisfactory. Instead, we use an area-weighted average of the six ocean gridboxes, which cover the area of each atmosphere gridbox, making full use of all the information available.

The information passed from the atmosphere to the ocean describes the surface fluxes of momentum, heat and fresh water. It is essential that the coupling should be conservative i.e. the area-average of each quantity on the two grids must be equal. We could set the six ocean points in an atmosphere box all equal to the single atmosphere value for the box, but we prefer to have ocean fields which vary continuously in space. To achieve both these aims, we first use bilinear interpolation from the atmosphere grid to obtain a field which varies on the higher-resolution ocean grid. Then we calculate the area-average of the six ocean boxes in each atmosphere box, and compare it with the atmosphere value. Since interpolation is not conservative, the values will not generally be equal. We adjust the six ocean values by dividing by the ratio of the average to the atmosphere value, or subtracting the difference, to regain equality in each atmosphere box. Division is used for positive-definite quantities, such as downward penetrative shortwave radiation (subtraction might give unphysical negative values), and addition for others, such as net surface water flux (where there might be six nearly balancing values of opposite signs, and division by their small sum could produce unreasonably large values). This adjustment procedure depends on the coincidence of the ocean and atmosphere land-sea masks, so that both grids have the same ocean area in each atmosphere box.

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